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Avionics Project Based Learning

Numerical Modeling of Electromagnetic Scattering for Radar Cross Section Prediction

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INTRODUCTION

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Why Radar Cross Section (RCS) Matters?

- RCS is NOT the physical size of an object.
- It quantifies how effectively a target reflects electromagnetic energy back toward the radar receiver.
- A lower RCS directly reduces the radar detection range.

$$R_{\max} = \left(\frac{P_t G_t G_r \lambda^2 \sigma}{(4\pi)^3 S_{\min}} \right)^{1/4} \quad R_{\max} \propto \sigma^{1/4}$$



F-15 Eagle

Typical Frontal RCS

+10 to +14 dBsm



B-2 Spirit

Typical Frontal RCS

−20 to −30 dBsm

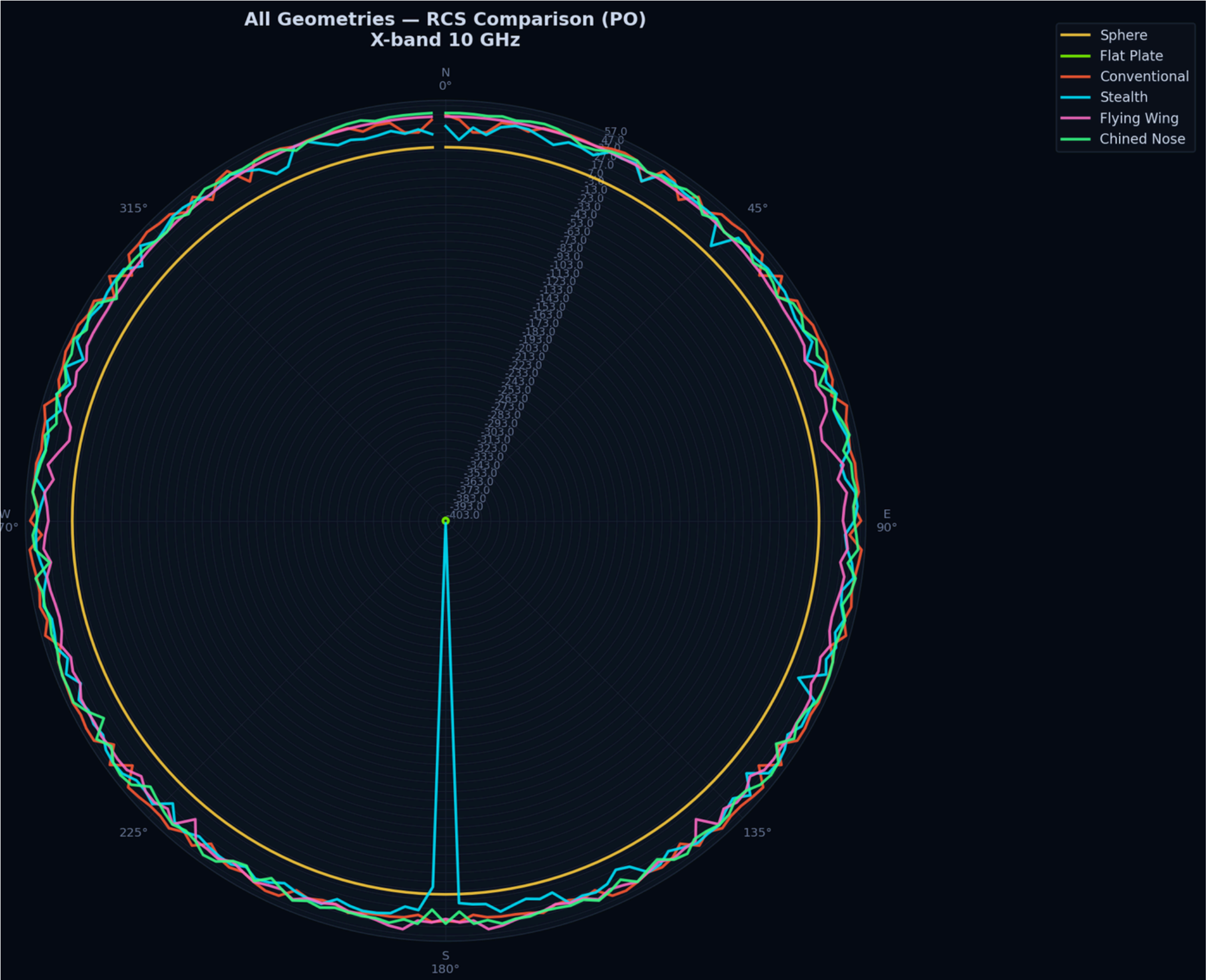
$$\sigma = \lim_{R \rightarrow \infty} 4\pi R^2 \frac{|E_s|^2}{|E_i|^2}$$

Equation Parameters:

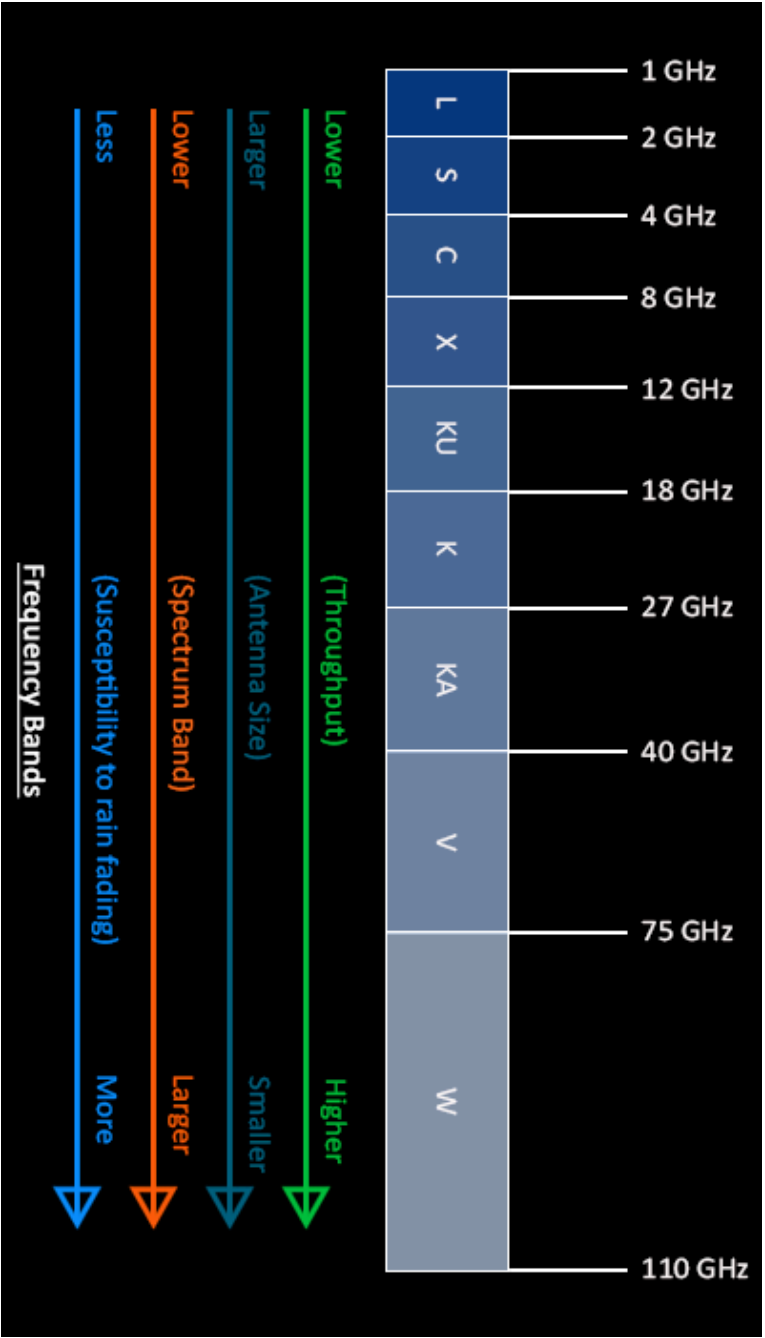
- P_t Transmitted Power
- G_t Transmitter Antenna Gain
- G_r Receiver Antenna Gain
- λ Radar Wavelength
- σ Radar Cross Section
- $S_{\min}$ Minimum Detectable Signal
- E_i Incident Electric Field
- E_s Scattered Electric Field
- R Distance to Observation Point

- Radar systems operate across L-band (1 GHz) to Ka-band (36 GHz).
- Electromagnetic scattering is analysed using three complementary mechanisms.
- Framework validated against the analytical Mie series solution.
- Six representative geometries are investigated.
- Outputs include polar plots, heatmaps, frequency sweeps and RAM analyses.
- Entire computational framework developed in Python.

PROJECT OVERVIEW



Comparison of RCS polar signatures for the six aerospace geometries analysed in this study.





Conventional aerospace structures incorporate geometric features that produce strong specular and edge-diffraction radar returns across multiple aspect angles:

Primary Scattering Contributors in Conventional Designs:

Feature	Geometry	Approximate RCS (X-band)
Vertical stabiliser (3 m × 2 m)	Flat-plate broadside	~+57 dBsm
Cylindrical fuselage (d=1.5 m, L=15 m)	Cylinder broadside	~+42.5 dBsm
Unshielded engine inlet	Deep cavity retroreflector	+15 to +25 dBsm
Wing–fuselage junction	Dihedral corner reflector	+10 to +20 dBsm
Canopy / cockpit	Cavity forward aspect	+5 to +15 dBsm



Literature Survey

Year	Author(s)	Journal / Publication	Key Technical Contribution	Relevance to Our Project
1908	Gustav Mie	Beiträge zur Optik trüber Medien	Developed the exact analytical solution (Mie Theory) for electromagnetic scattering by a sphere.	<i>Radar Cross Section Validation baseline for our computational RCS model.n (2nd ed.)</i>
1962	P. Ya. Ufimtsev	Method of Edge Waves in the Physical Theory of Diffraction	Introduced the Physical Theory of Diffraction (PTD) for modelling edge diffraction.	<i>Foundation for edge diffraction modelling in complex aircraft geometries.</i>
2004	E. F. Knott, J. F. Shaeffer & M. T. Tuley	Radar Cross Section (2nd Edition)	Comprehensive treatment of RCS theory, Physical Optics (PO), PTD, canonical targets and shaping methods.	<i>Primary reference for the computational methodology adopted in this project.</i>
2001	M. I. Skolnik	Introduction to Radar Systems (3rd Edition)	Radar equation, radar frequency bands, target detection fundamentals and radar system performance.	Used for radar fundamentals and interpretation of RCS behaviour.



Modern radar systems remain a primary means of detecting, tracking, and identifying aerospace vehicles. RCS is strongly dependent on external geometry, orientation, and frequency. Traditional aerospace configurations, designed primarily for aerodynamic, structural, mission requirements, produce geometric features that increase detectability. Stealth technology addresses this through geometric shaping that redirects scattered energy away from the radar receiver.

Challenge 1: Three physically distinct scattering mechanisms (specular, edge, multi-bounce) must be modelled coherently, with phase.

Challenge 2: RCS depends non-linearly on surface orientation, edge wedge angle, and dihedral geometry.

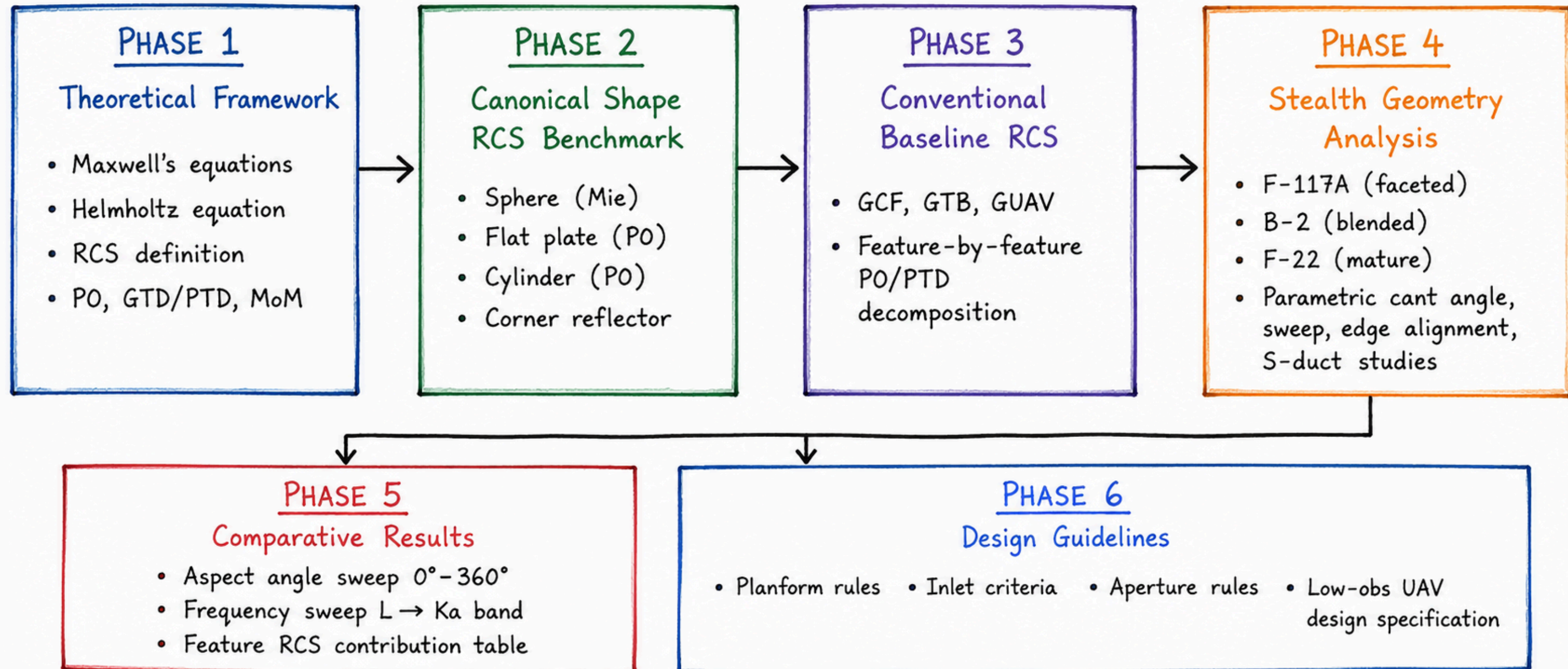
Challenge 3: Which geometric features drive which mechanism and by how much, must be extracted quantitatively from results, not assumed.



- Design a first-principles RCS computation framework in Python integrating Physical Optics, Physical Theory of Diffraction, double-bounce, and RAM models.
- Develop and validate the Physical Optics solver against the exact Mie series solution for a conducting sphere.
- Analyse the RCS of six triangulated-mesh geometries across azimuth, elevation, and frequency.
- Evaluate the dominant scattering mechanisms across geometries and derive low-observable design principles.



METHODOLOGY FLOWCHART





Six triangulated-mesh geometries, all PEC, coarse meshing (18–34 facets) for rapid comparative sweeps:

- Sphere ($r=1\text{m}$, 2304 facets) - Mie validation baseline
- Flat plate ($2\text{m}\times 2\text{m}$, 2 facets) - textbook $\sigma=4\pi A^2/\lambda^2$ check
- Conventional aircraft (34 facets) - flat fuselage, right-angle wing/fin dihedrals, flat nose-on intake face
- Stealth aircraft (20 facets) - trapezoidal fuselage, pyramidal nose, 65° swept delta wings, 35° -canted V-tail
- Flying wing (28 facets) - zero vertical surfaces, sawtooth trailing edge
- Chined nose (18 facets) - diamond fuselage blended to wing LE, 28° twin tails

Scattering Mechanisms:

Mechanism	Physical Description
Physical Optics (PO)	Specular reflection from illuminated facets aligned with the incident wave
Physical Theory of Diffraction (PTD)	Edge diffraction from geometric discontinuities (wedges, edges)
Double-Bounce	Second-order retroreflection from two facets forming a corner reflector



Physical Optics (PO) — Specular Reflection

$$f_{PO} = \sum_i \rho_i A_i (\hat{n}_i \cdot \hat{s}) \exp(j2k \mathbf{c}_i \cdot \hat{s})$$

PO specular return scales with facet area and incidence angle as:

$$\sigma \propto \frac{A^2 (\hat{n} \cdot \hat{s})^2}{\lambda^2}$$

Physical Theory of Diffraction (PTD) — Edge Diffraction

$$D(n, \beta) = -\frac{e^{-j\pi/4}}{n\sqrt{2\pi k}} \cdot \frac{\cot\left(\frac{\pi}{2n}\right)}{\sin \beta}$$

Diffraction lobes from finite edges of length L are governed by:

$$f_{PTD} \propto \text{sinc}(kL \cos \beta)$$



4 Double-Bounce (Corner Reflector) Condition

A second-order double-bounce contribution is active when the retroreflection condition is satisfied for a facet pair j :

$$\mathbf{d}_{ref,j} \cdot \hat{\mathbf{s}} > 0.2 \quad (6)$$



Parameter	Value
Sphere radius	1 m
Frequency	10 GHz
ka	209.4 (optical regime)
Mie exact σ	4.97 dBsm
Optical limit (πr^2)	$3.14 \text{ m}^2 = 4.97 \text{ dBsm}$
PO solver match	Within 0.2 dBsm
Wiscombe convergence	n_max = 238 terms

Results 1: Validation & Physics-Level Effect

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PO-only computation underestimates RCS, missing edge and multi-bounce contributions present in real geometries.

Adding PTD raises the mean RCS by capturing edge diffraction from geometric discontinuities.

Adding double bounce introduces distinct spikes at 45°/135° quarter-on aspects, driven by 48 active retroreflecting facet pairs in the conventional aircraft.



Results 3: Edge and Double- Bounce Analysis

Edge Classification Comparison

Geometry	Total Edges	Internal Edges	Boundary Edges	Mean n
Conventional aircraft	61	41	20	1.8
Stealth aircraft	52	8	44	2

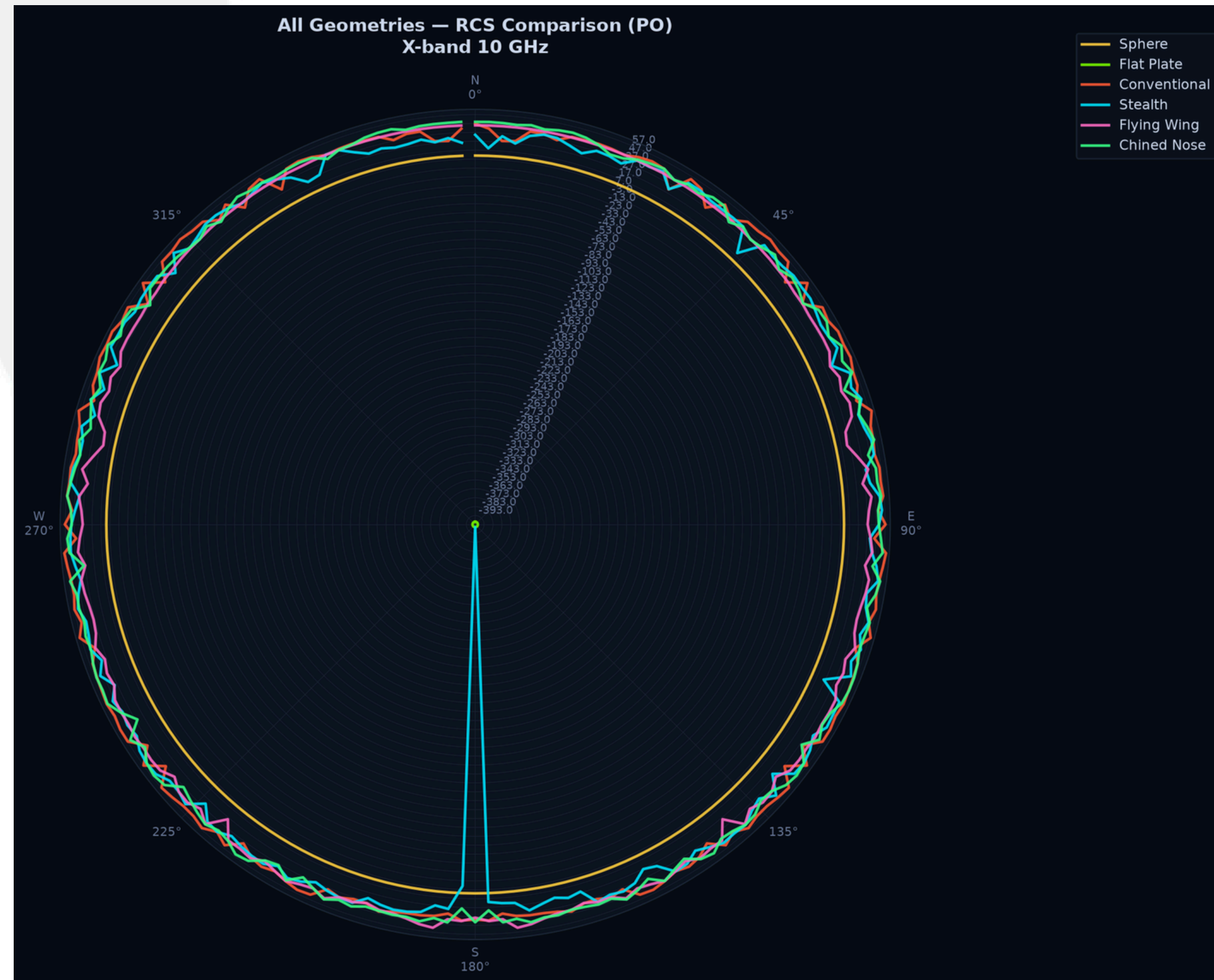
The conventional aircraft's 41 internal edges (many at $n=1.5$, right-angle convex ridges) drive elevated diffraction; the stealth aircraft's uniform $n=2.00$ reflects zero internal right-angle ridge structure.

Double-bounce hotspot matrix: conventional aircraft shows 48 active retroreflecting facet pairs clustered at the fin-fuselage 90° dihedral; stealth aircraft shows ~ 22 pairs at roughly half peak amplitude with no dominant cluster, as the 35° -canted V-tail eliminates the dominant corner reflector.



Results-MIE validation for sphere

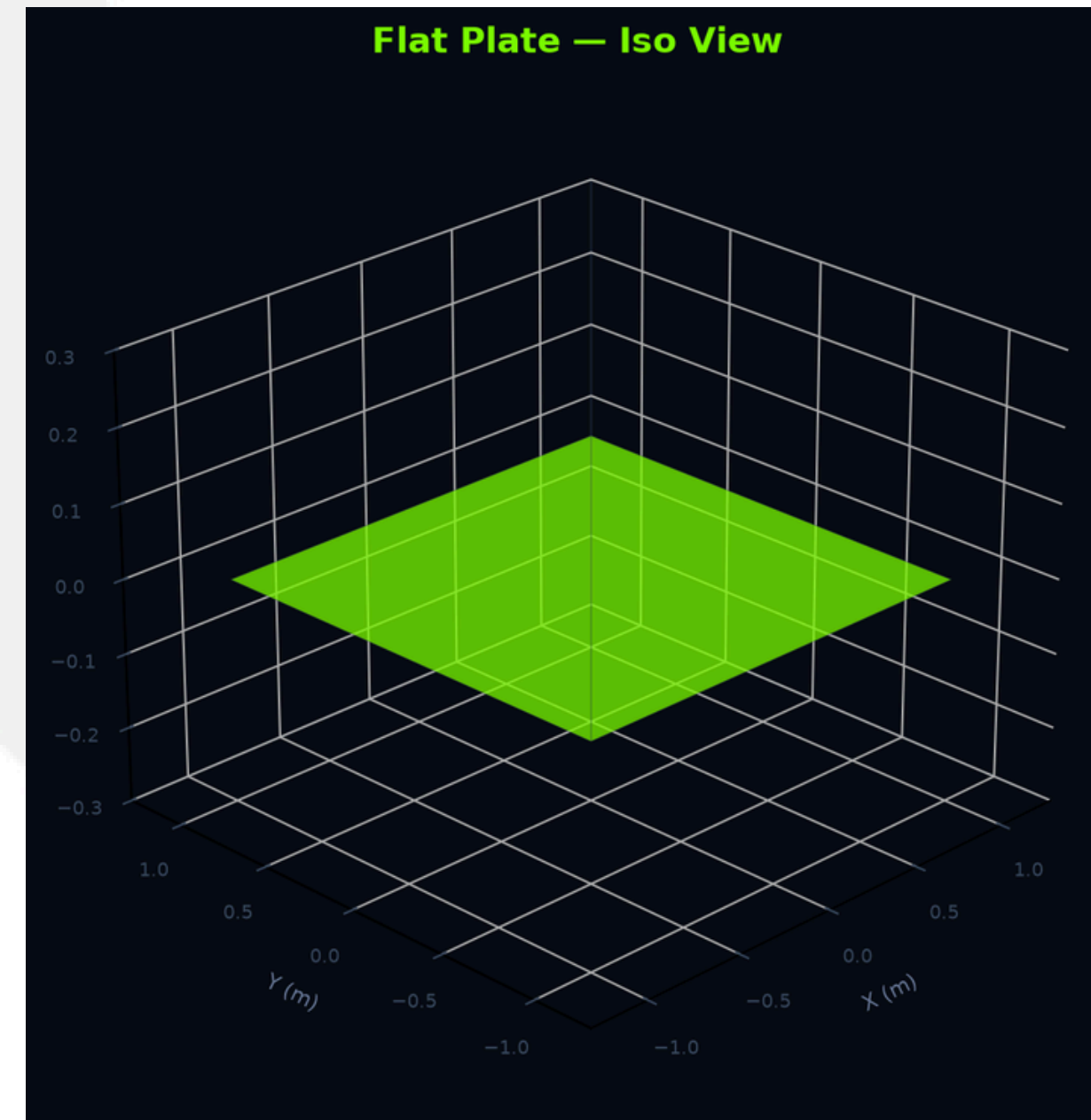
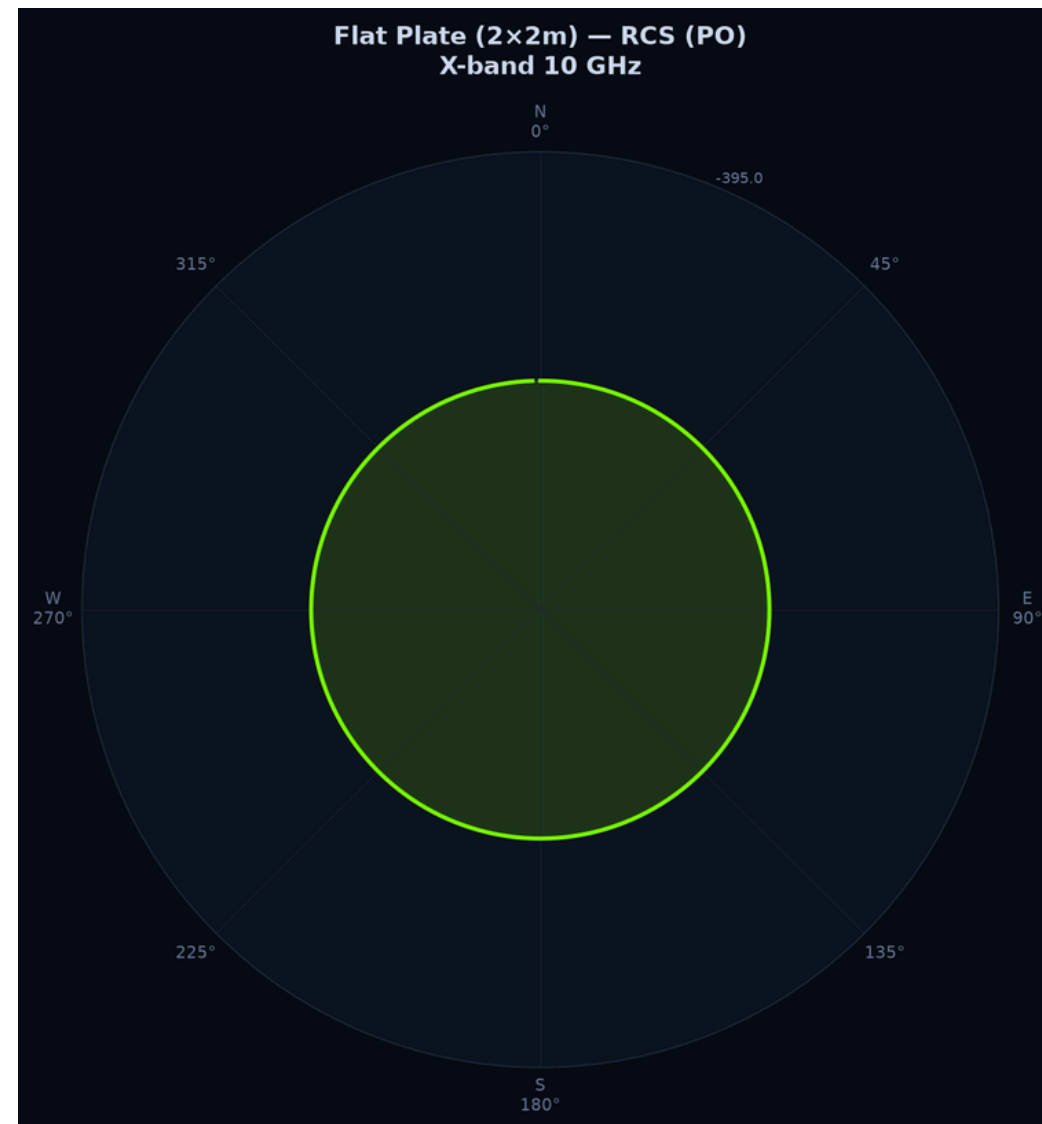
- The Sphere trace forms a smooth, near-perfect circle across all aspect angles, consistent with the isotropic scattering behavior predicted by Mie theory — confirming the PO solver is correctly calibrated against the canonical analytical benchmark.
- This circular, angle-independent response validates that the simulation setup (frequency, polarization, mesh resolution) reproduces the expected Mie series RCS for a sphere at X-band (10 GHz).
- The large dynamic range on the radial (dB) axis — from roughly -39 dB up to 57 dB — highlights how dramatically RCS varies with both geometry and aspect angle, which is the central message of a stealth-shaping study.





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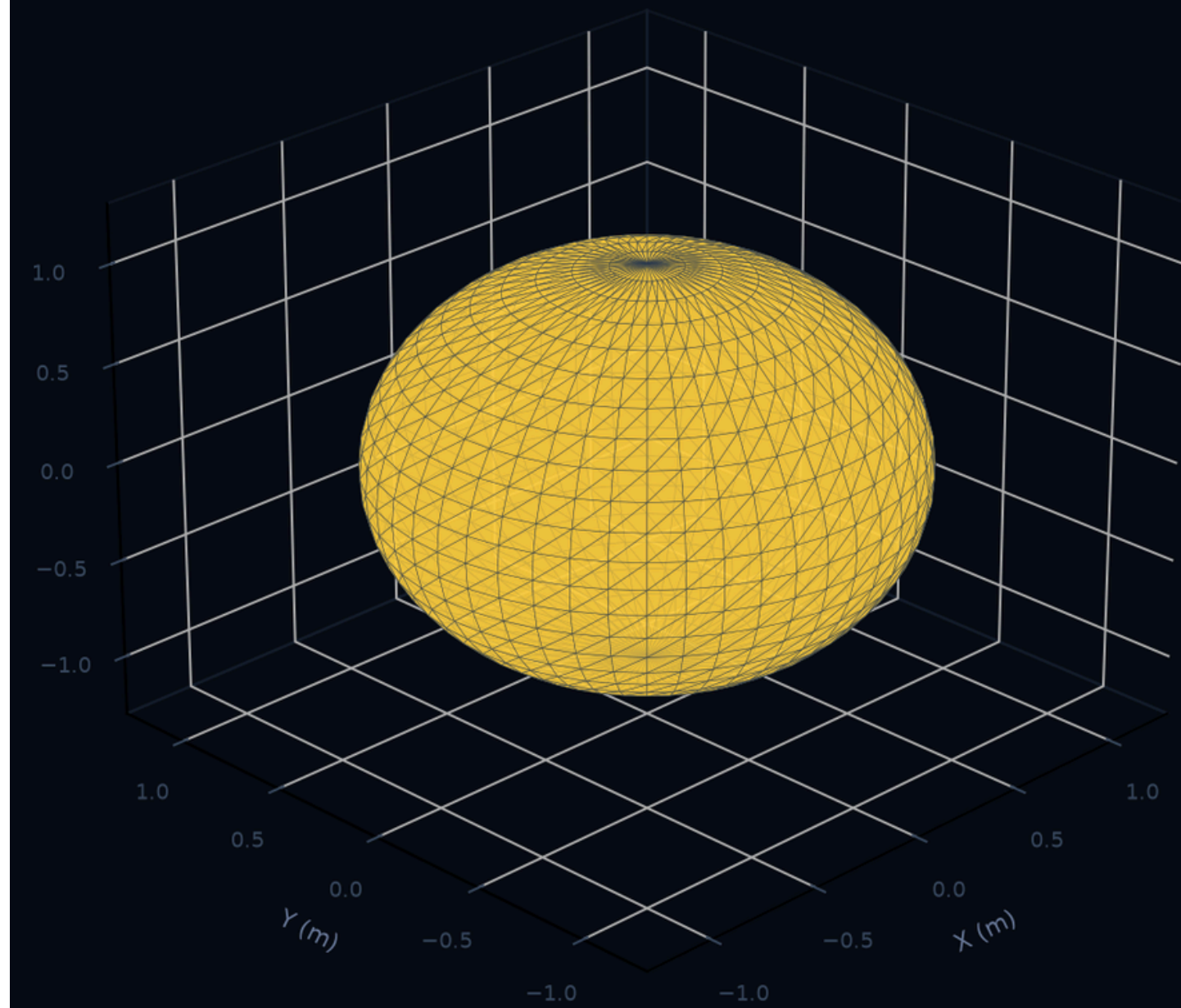




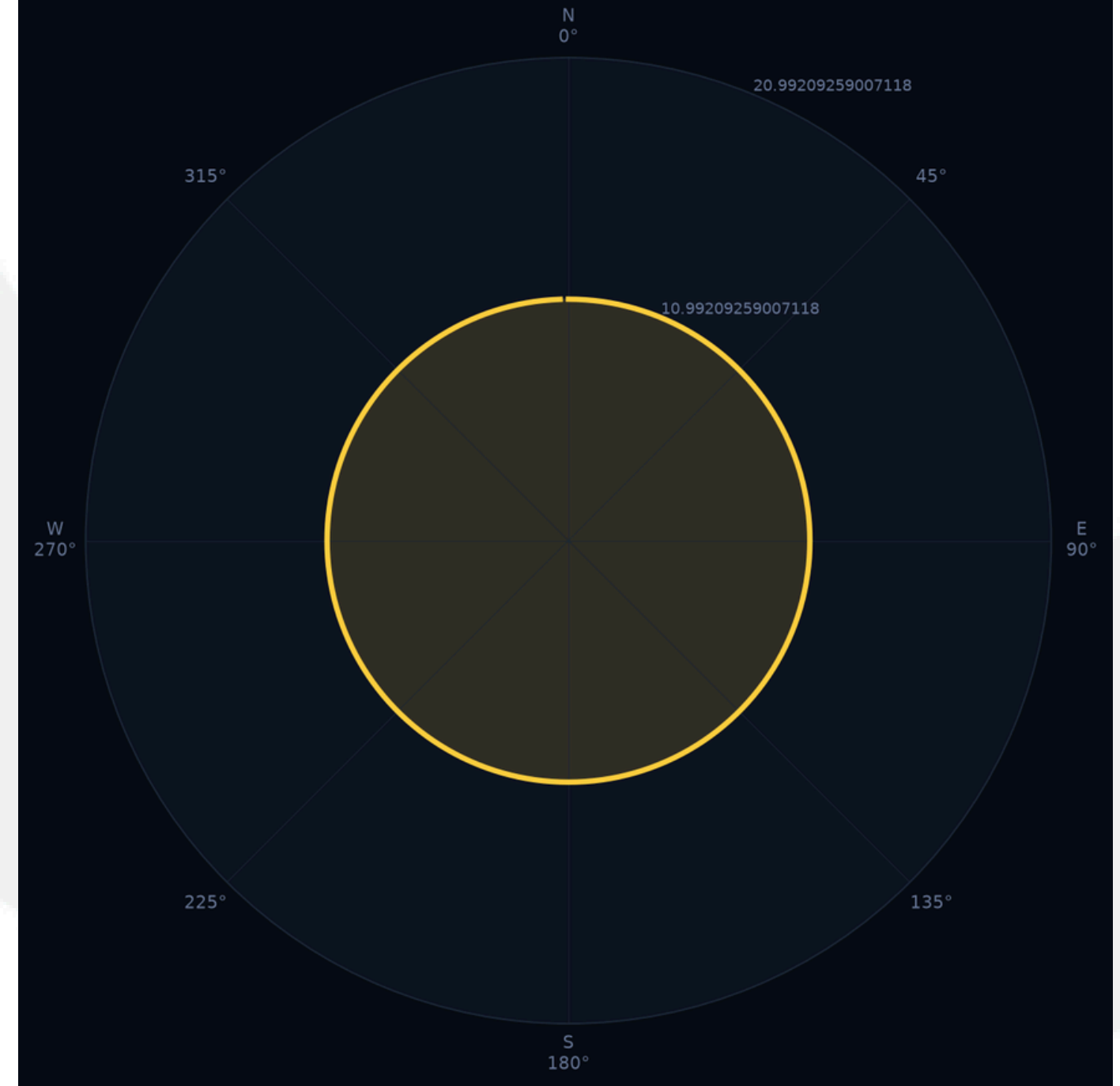
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Sphere — Iso View



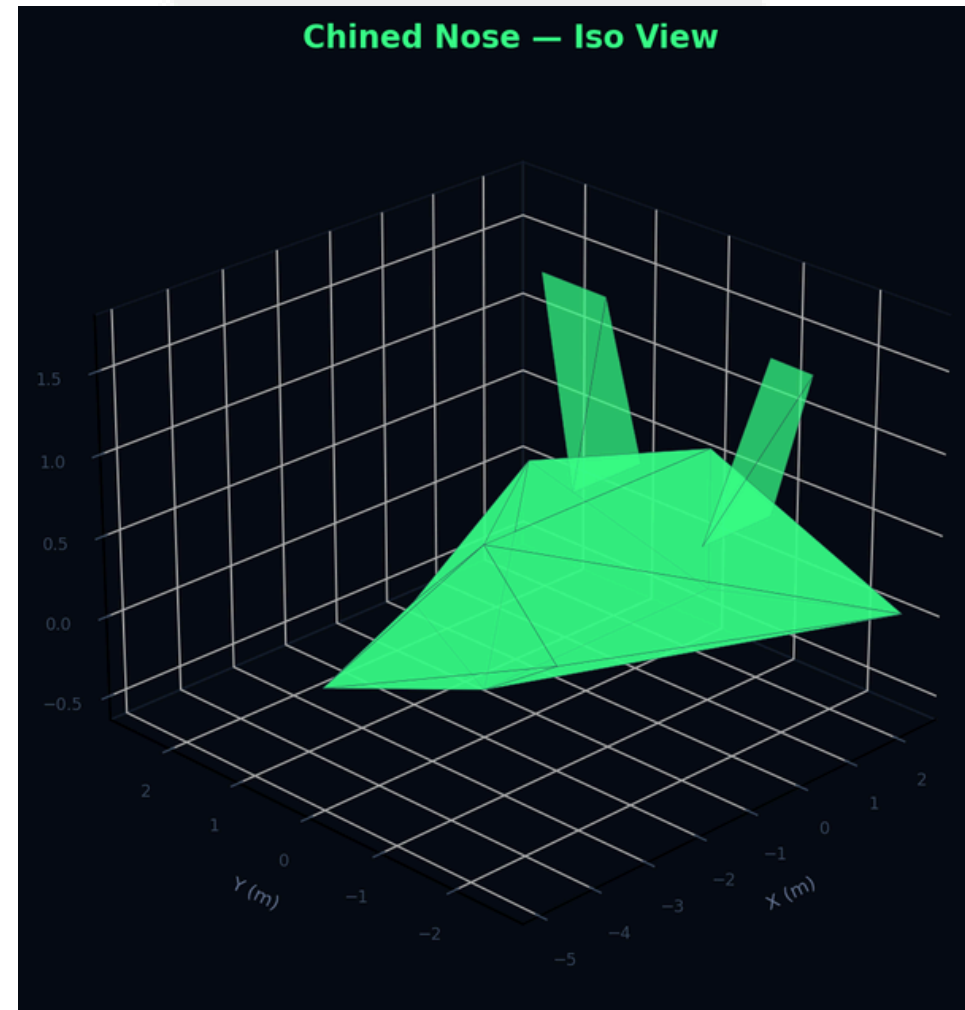
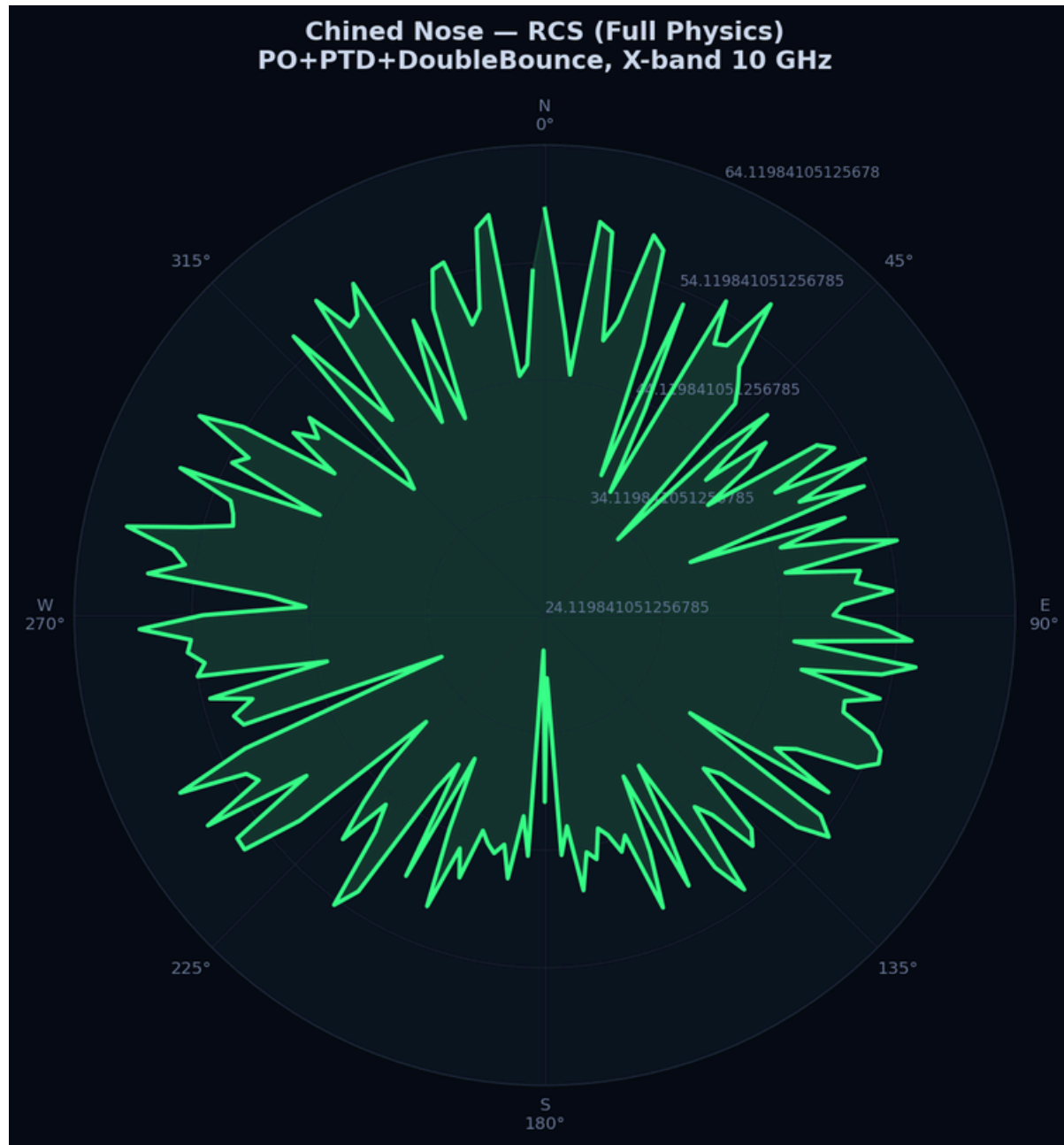
**Sphere (r=1m) — RCS (Mie Series, exact)
X-band 10 GHz**





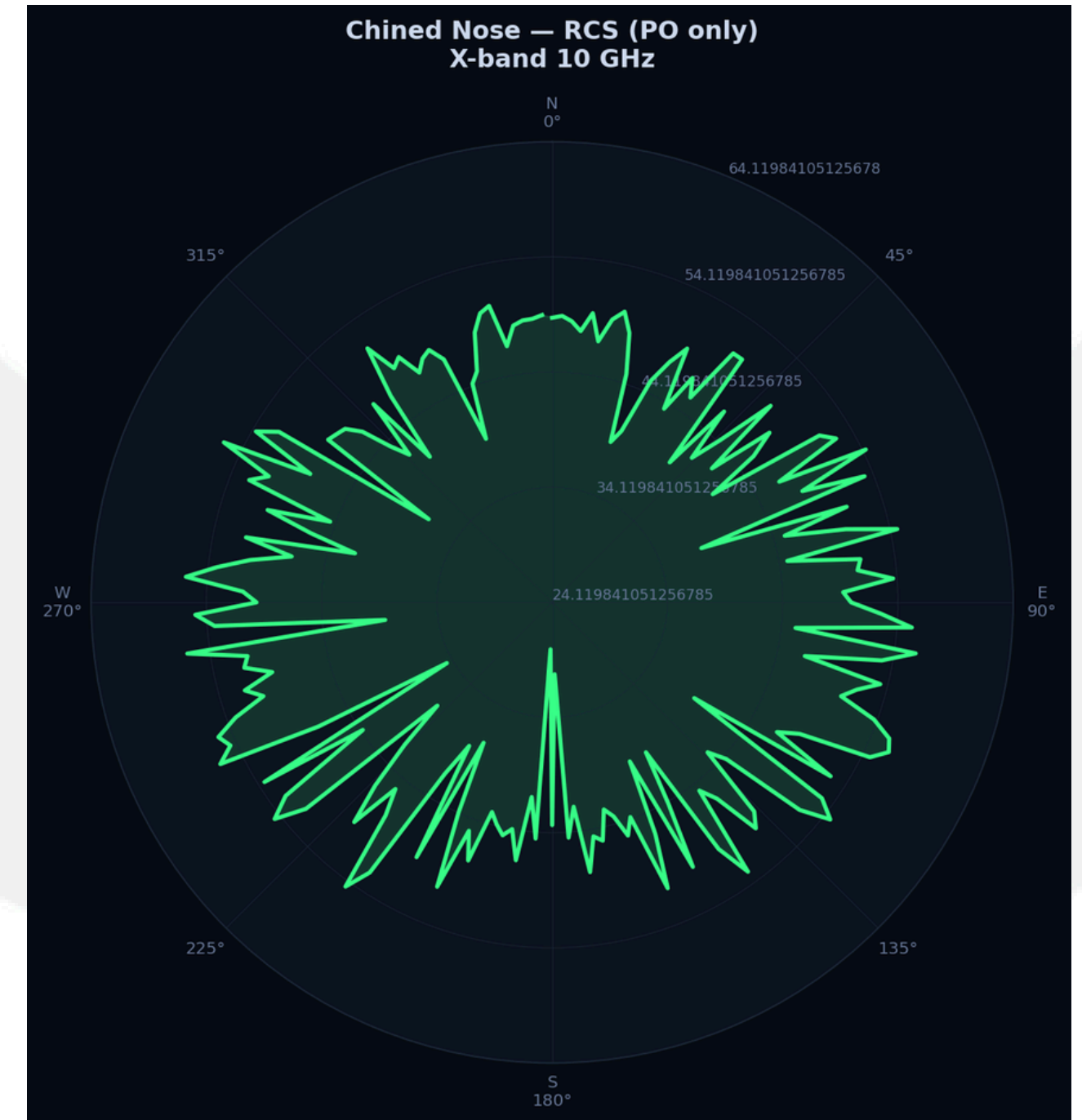
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Chined Nose - PO + PTD + DB



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Chined Nose - PO



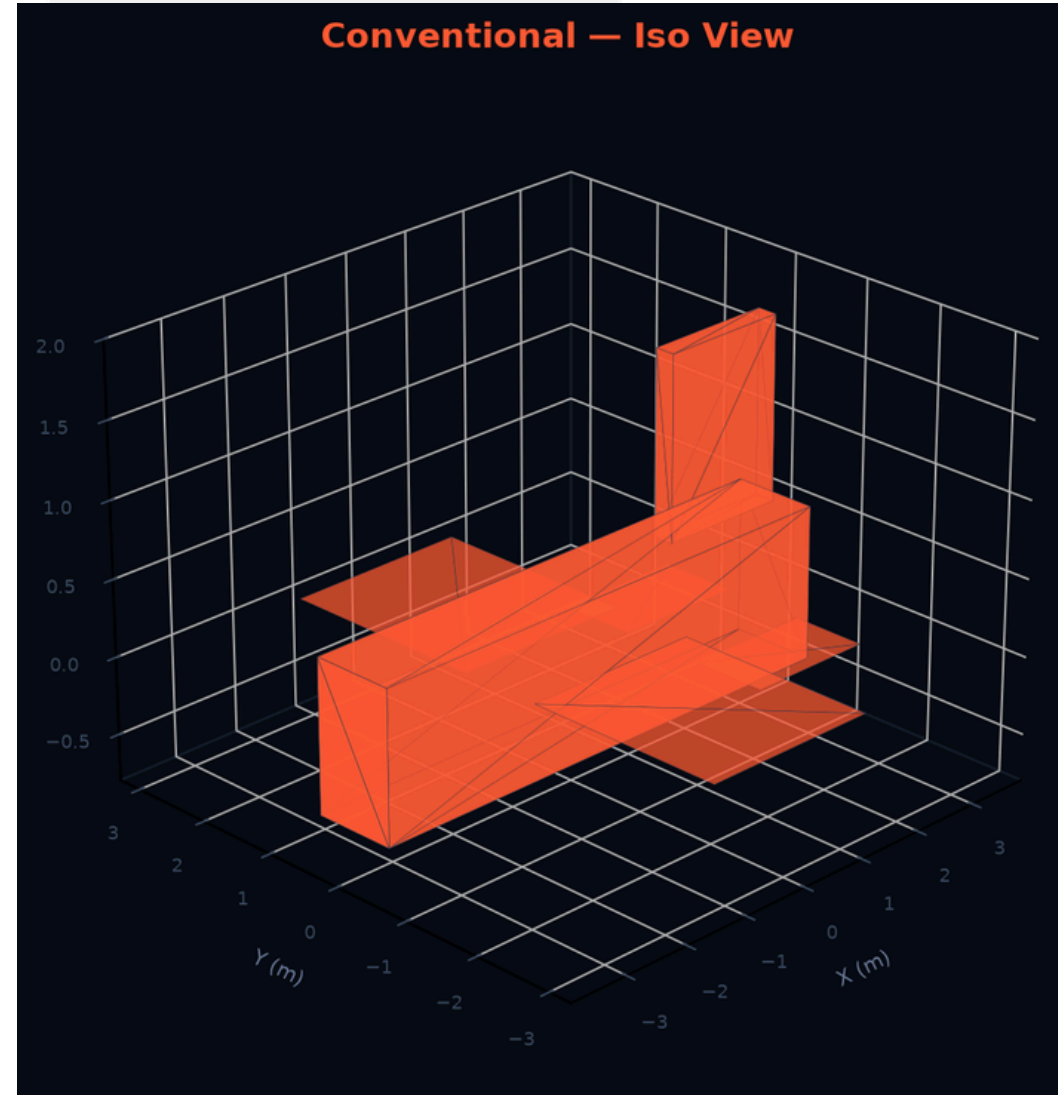


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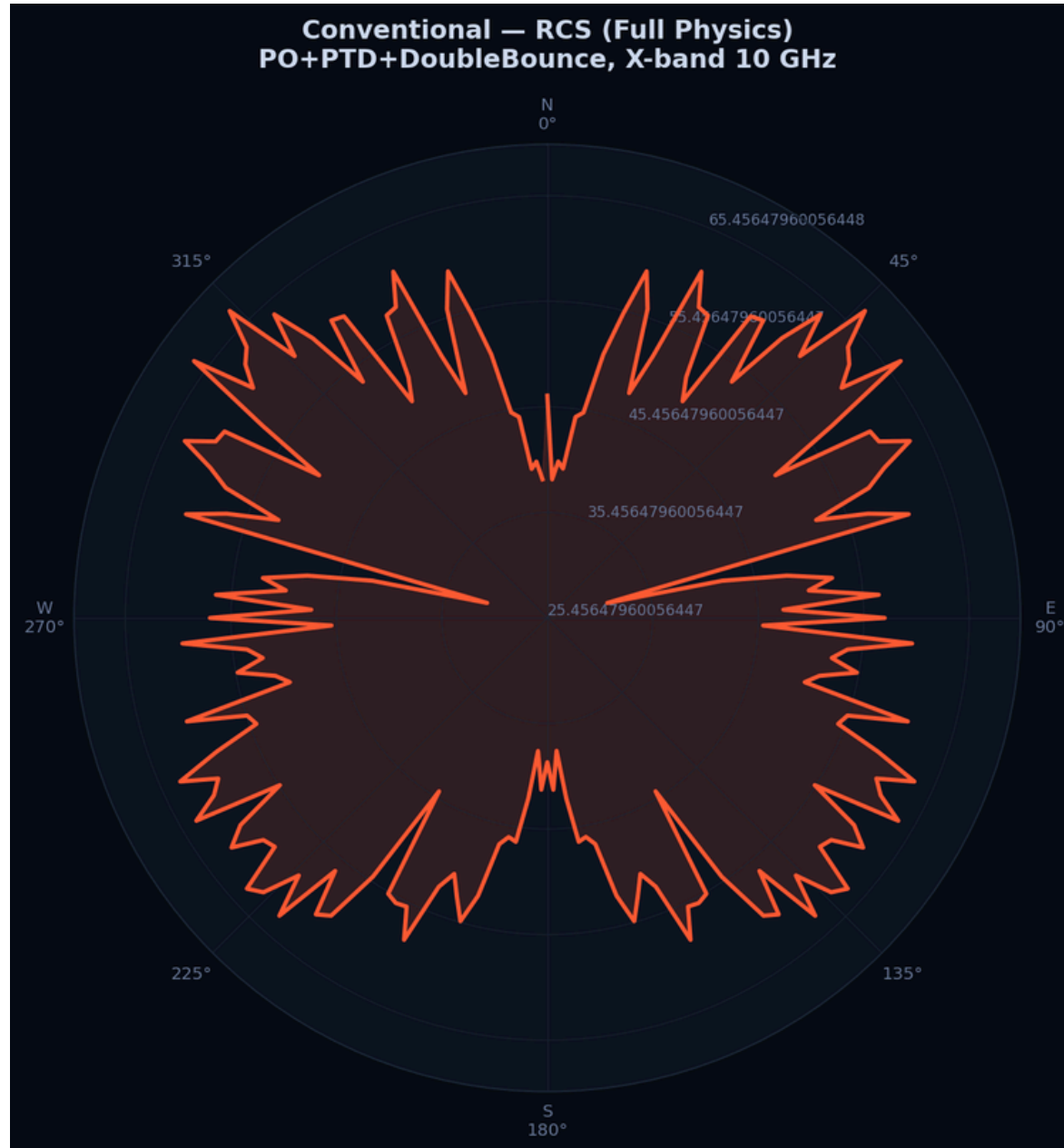
Convention

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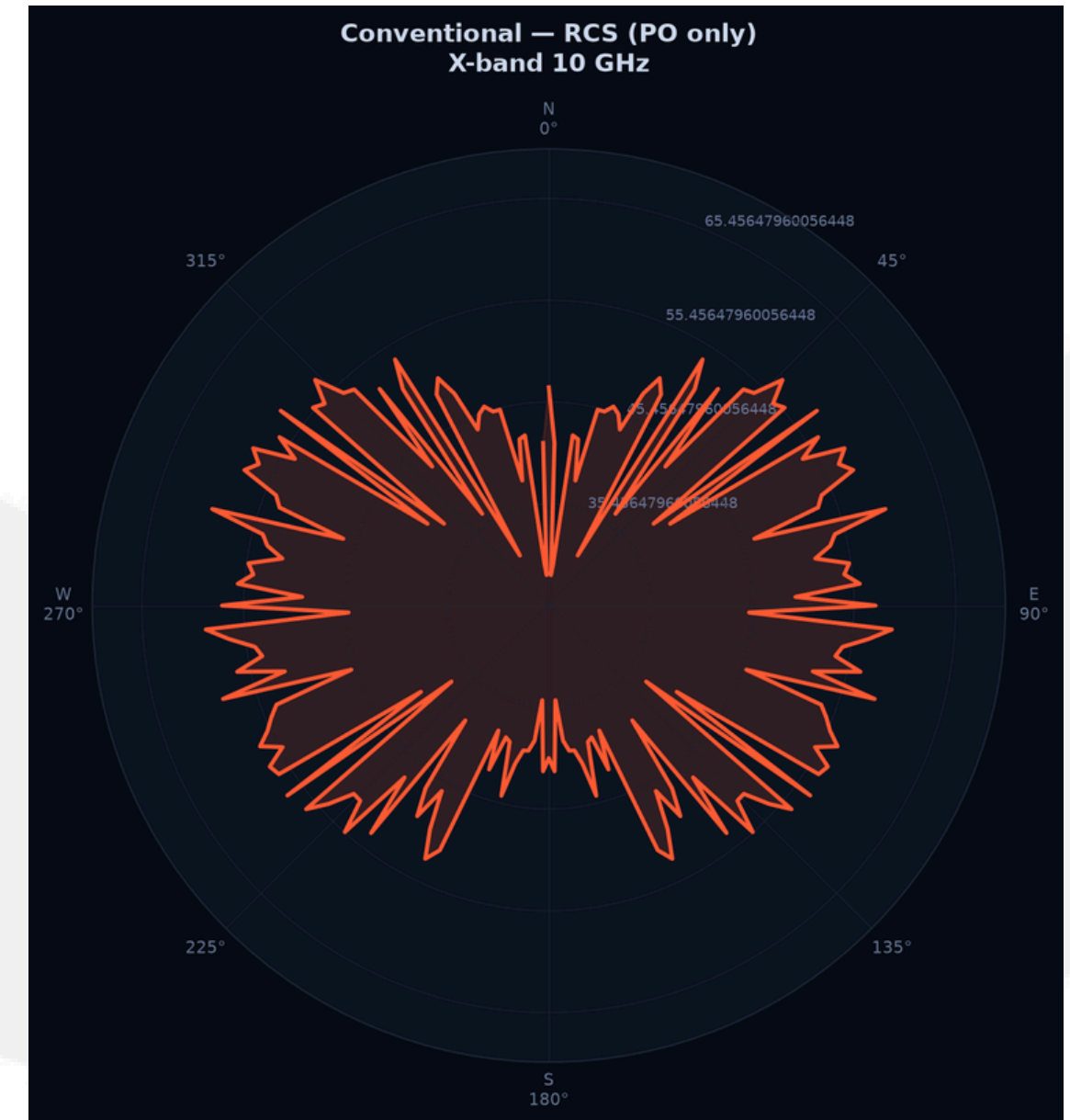
Conventional — Iso View



Conventional — RCS (Full Physics)
PO+PTD+DoubleBounce, X-band 10 GHz



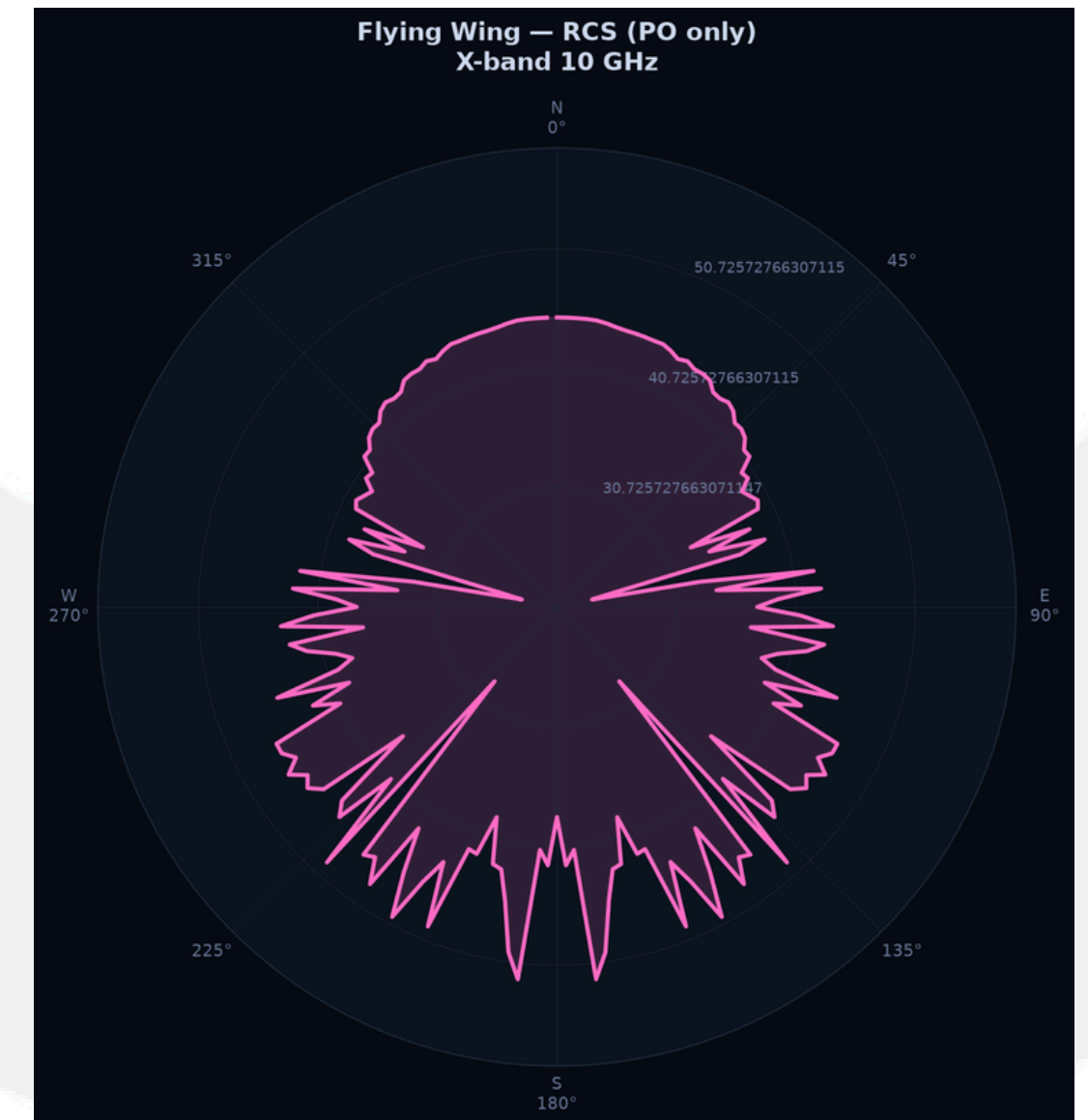
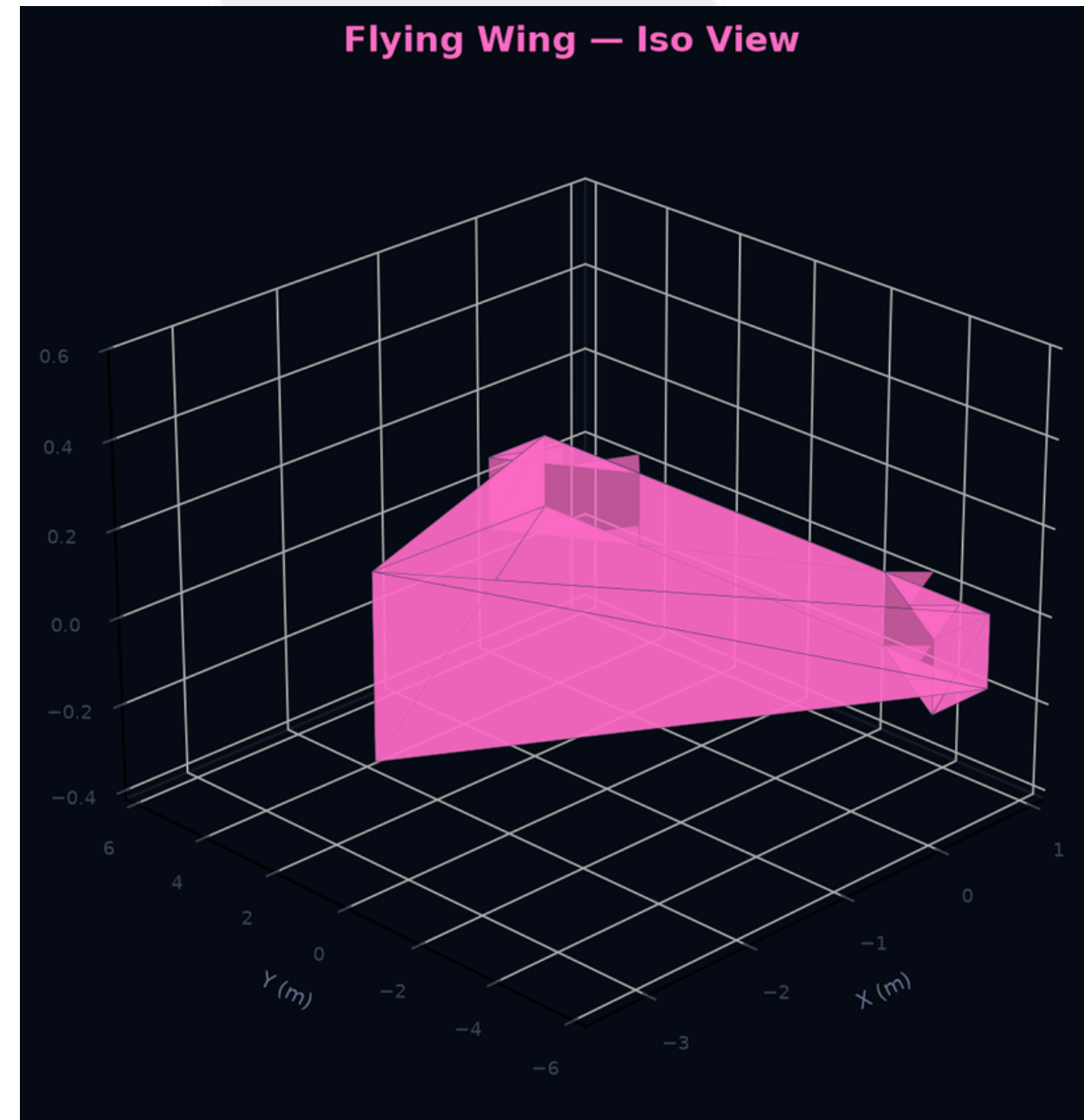
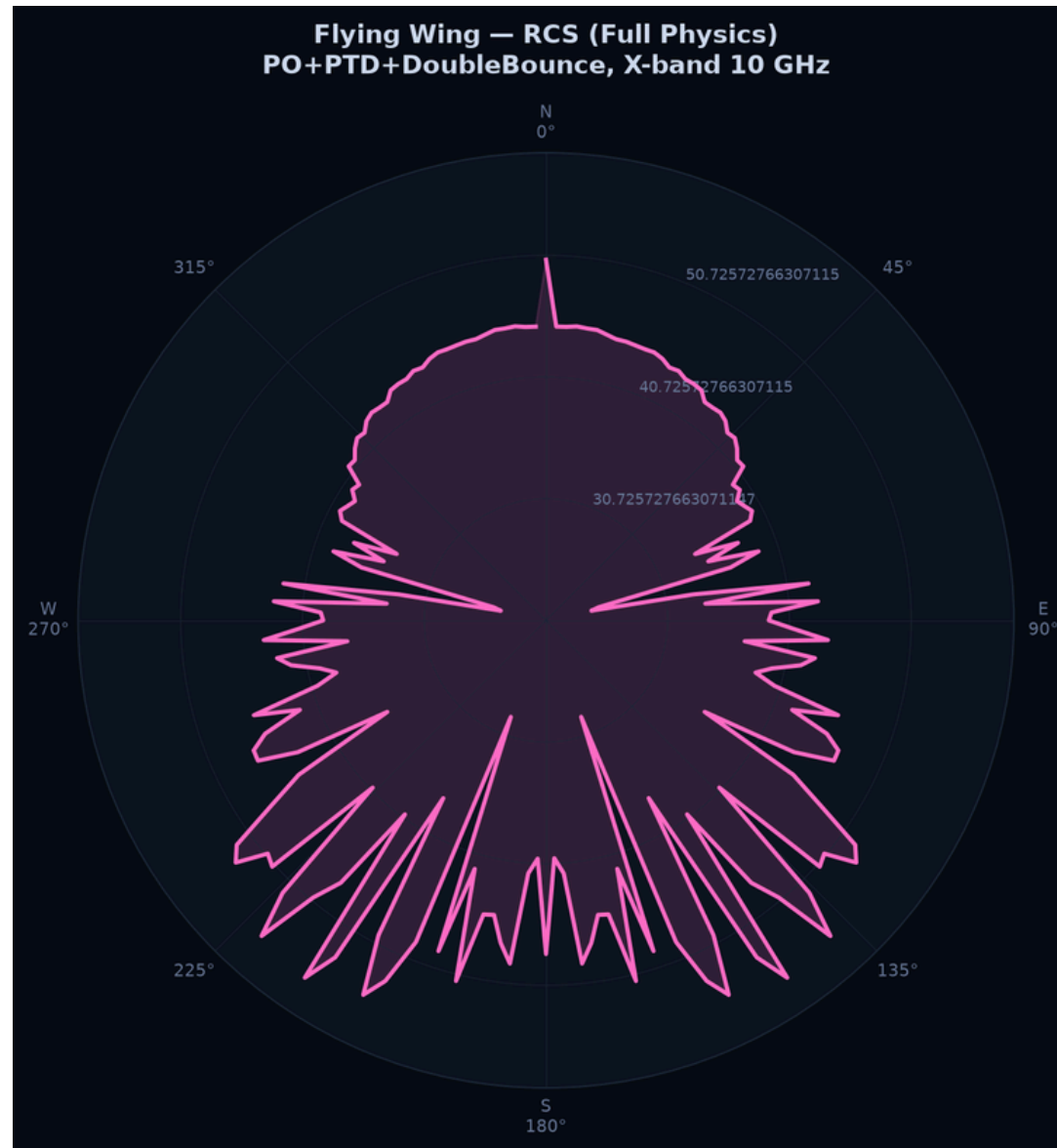
Conventional — RCS (PO only)
X-band 10 GHz

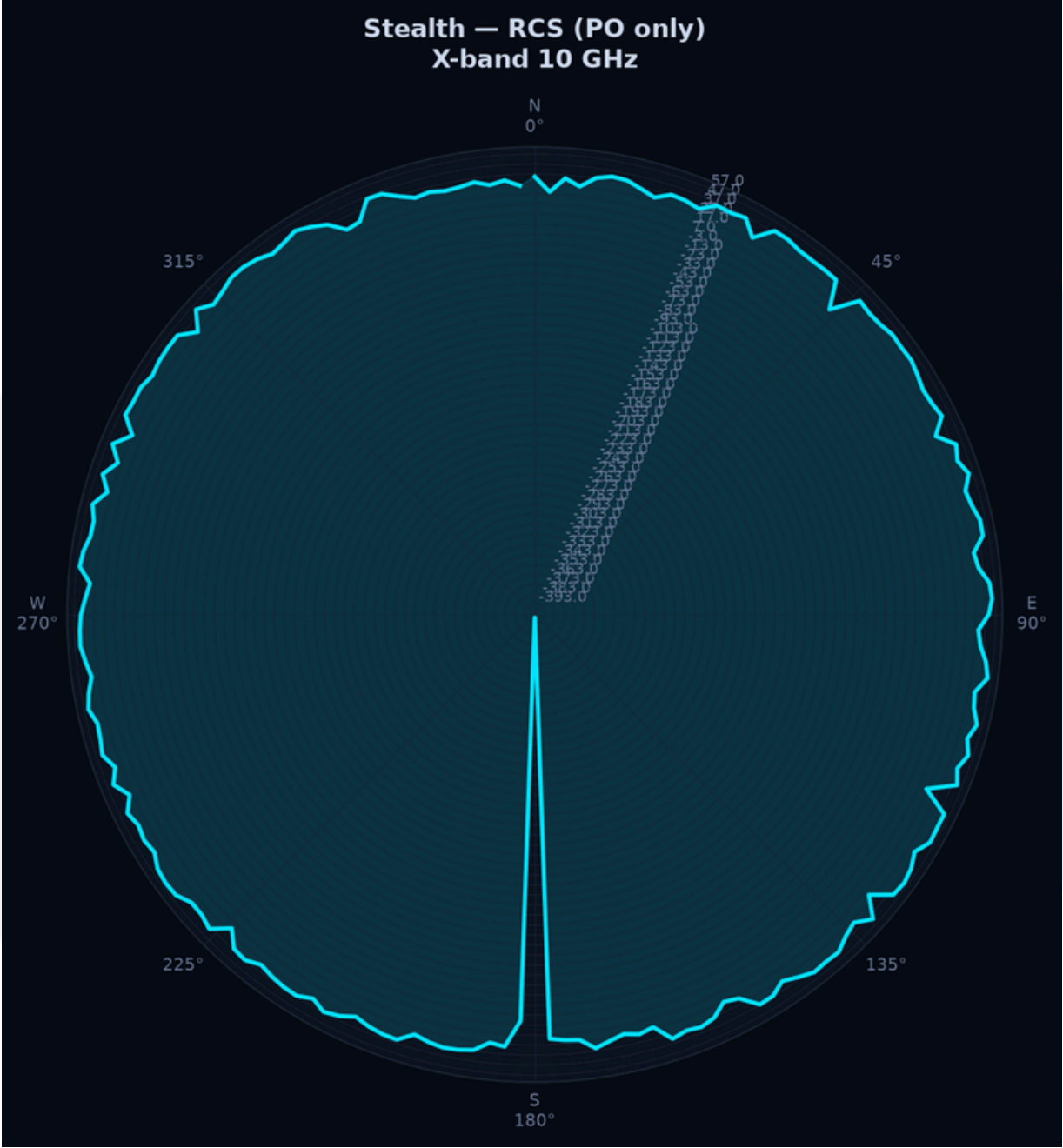
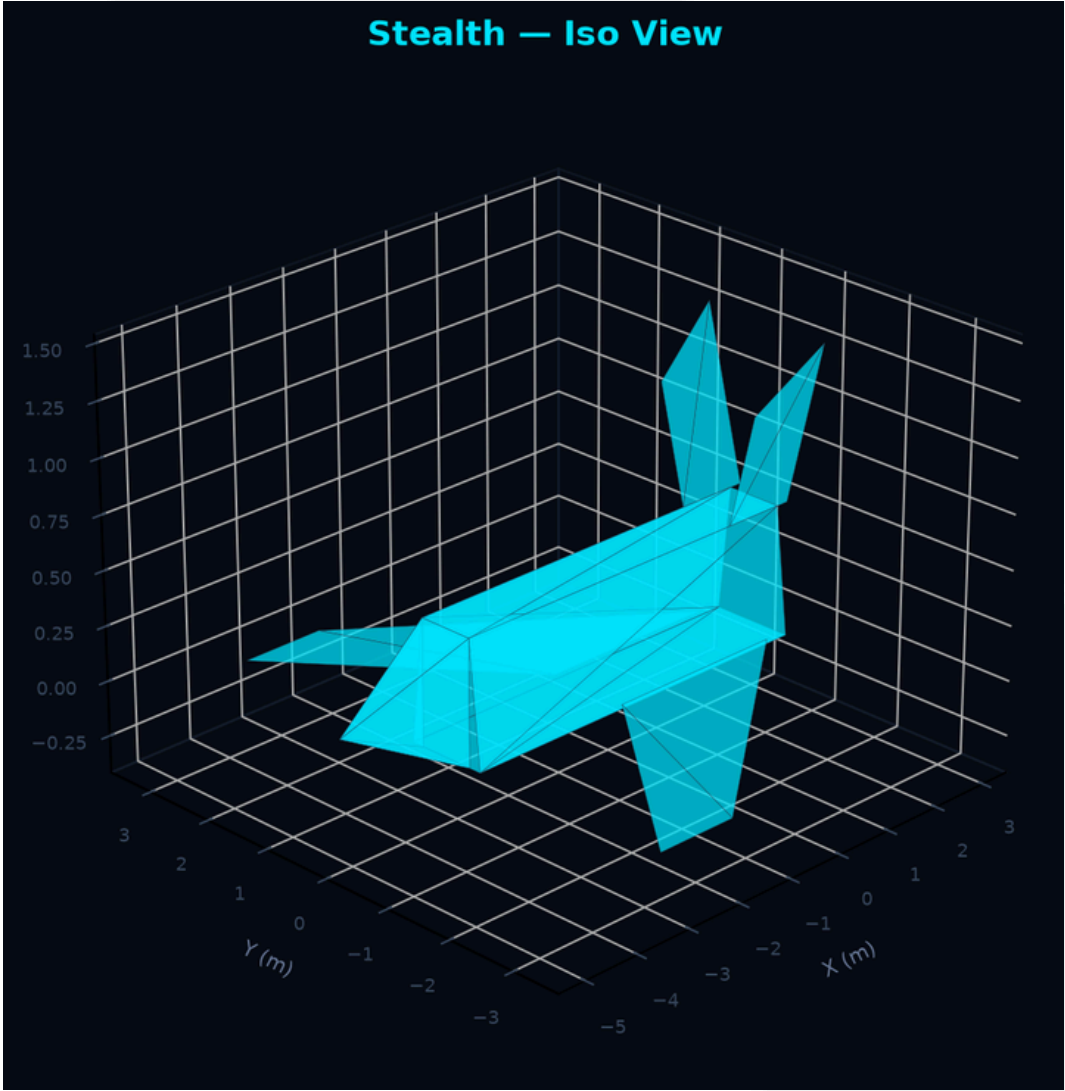
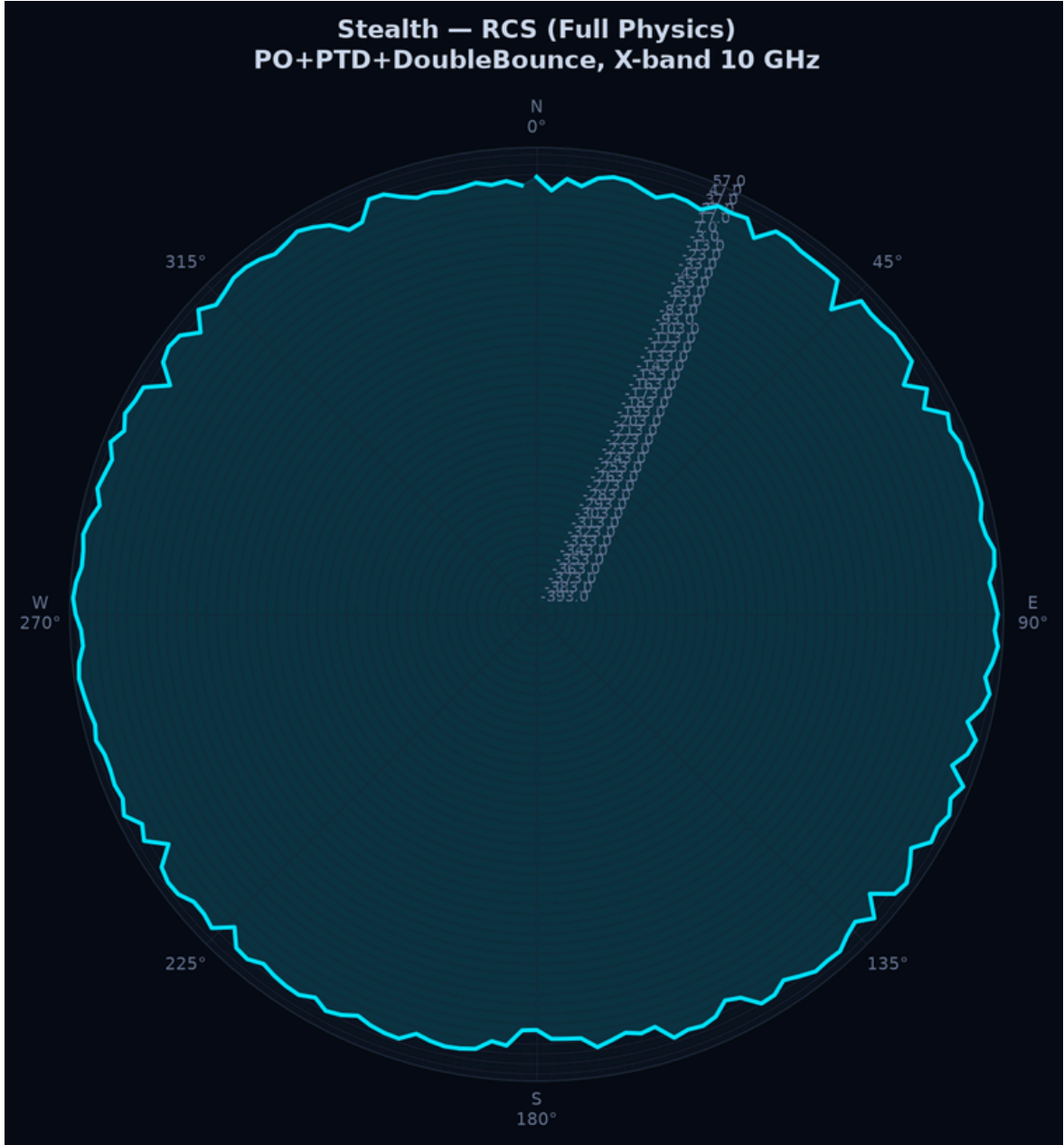




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Mean RCS Ranking (X-band, ascending)

Geometry	Mean RCS (dBsm)	Facet Count
Sphere	11	2304
Stealth aircraft	37.4	20
Flying wing	41	28
Chined-nose fighter	47	18
Conventional aircraft	49.5	34
Flat plate	53.5	2

Results 2: Geometry Comparison

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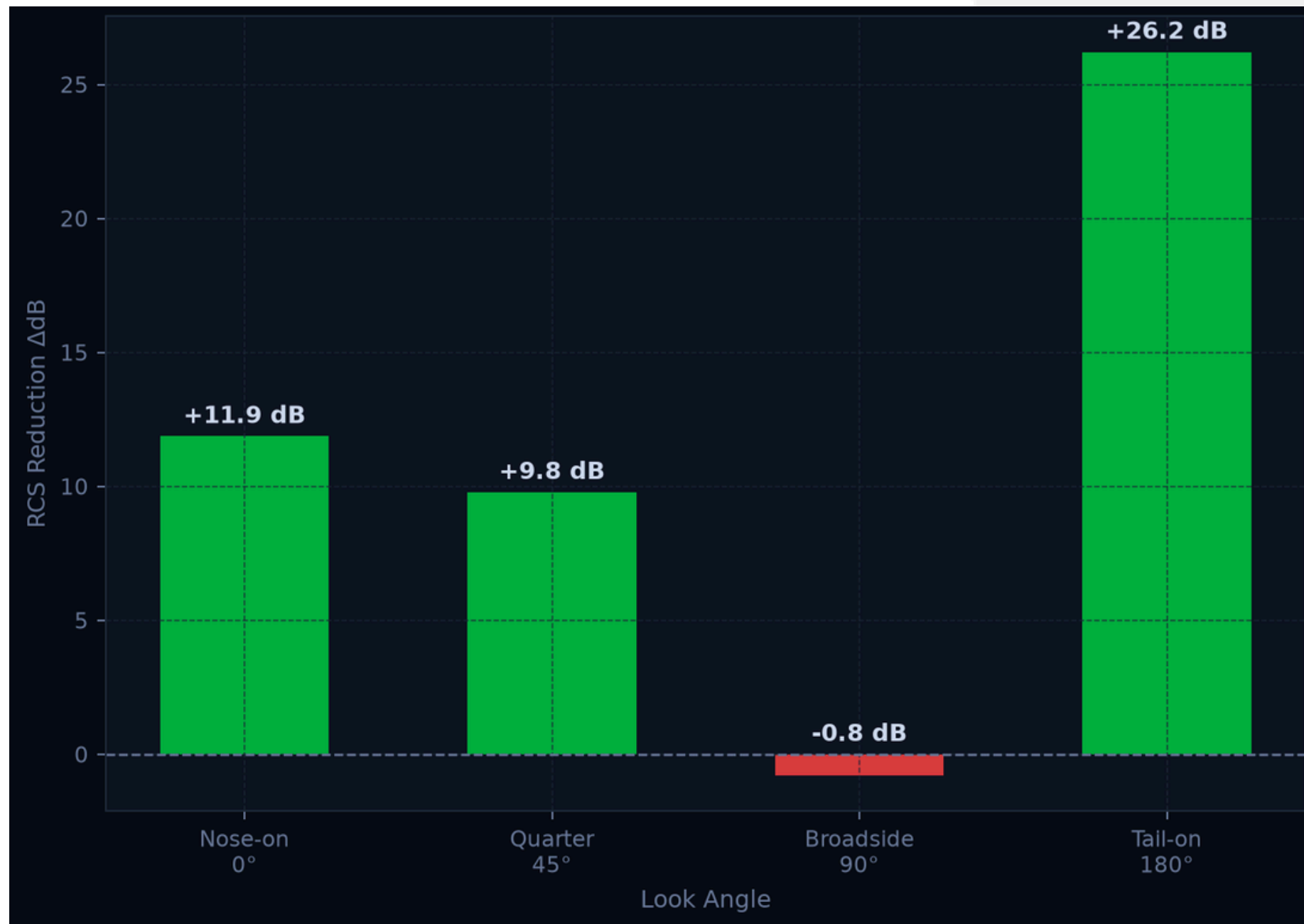
RCS Reduction: Conventional → Stealth

Aspect	Angle	Reduction (dB)
Nose-on	0°	11.9
Quarter-on	45°	9.8
Broadside	90°	-0.8
Tail-on	180°	26.2



RESULTS - RCS reduction: Conventional → stealth

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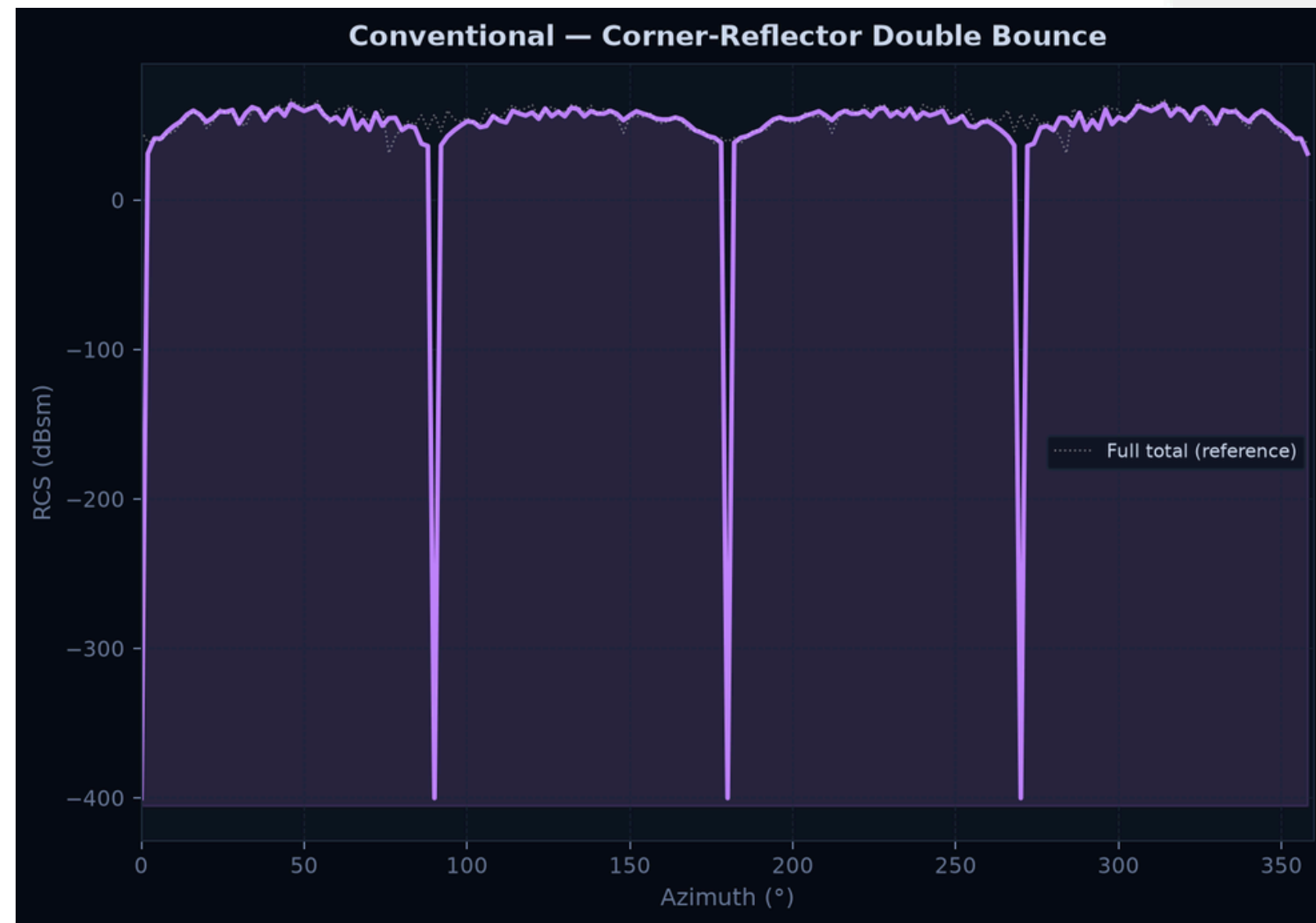
- **Nose-on (0°): +11.9 dB** — the mission-critical number. A frontal radar now only sees 6.5% of the energy it would from a conventional aircraft.
- **Quarter-on (45°): +9.8 dB** — corner reflector suppressed.
- **Broadside (90°): -0.8 dB** — near-zero, the V-tail cant partially re-radiates here.
- **Tail-on (180°): +26.2 dB** — the biggest win, pointed tail vs. flat engine face.



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RESULTS- Comparision between Conventional and Stealth geometry.

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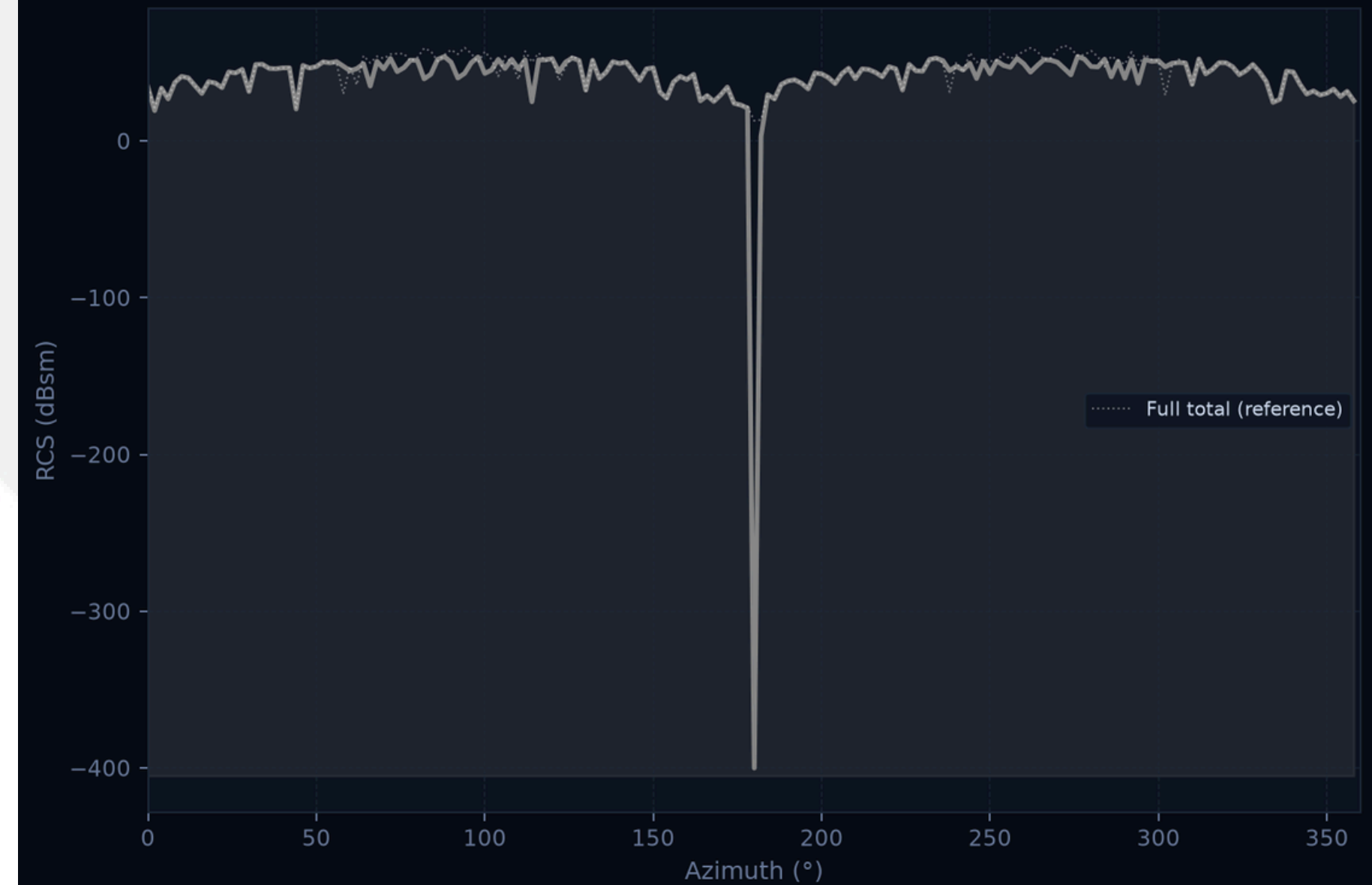
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Conventional — Specular Reflection (PO)



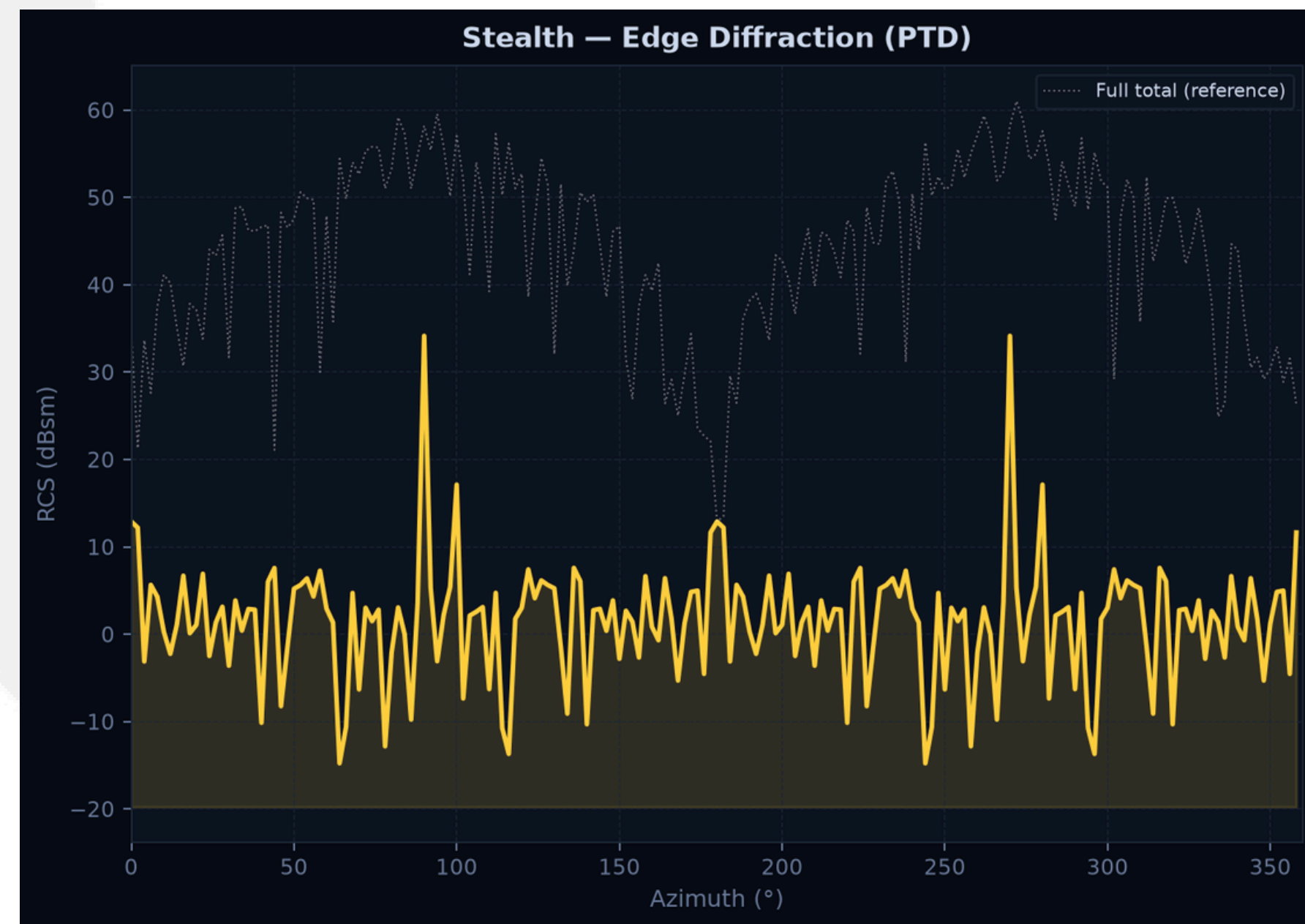
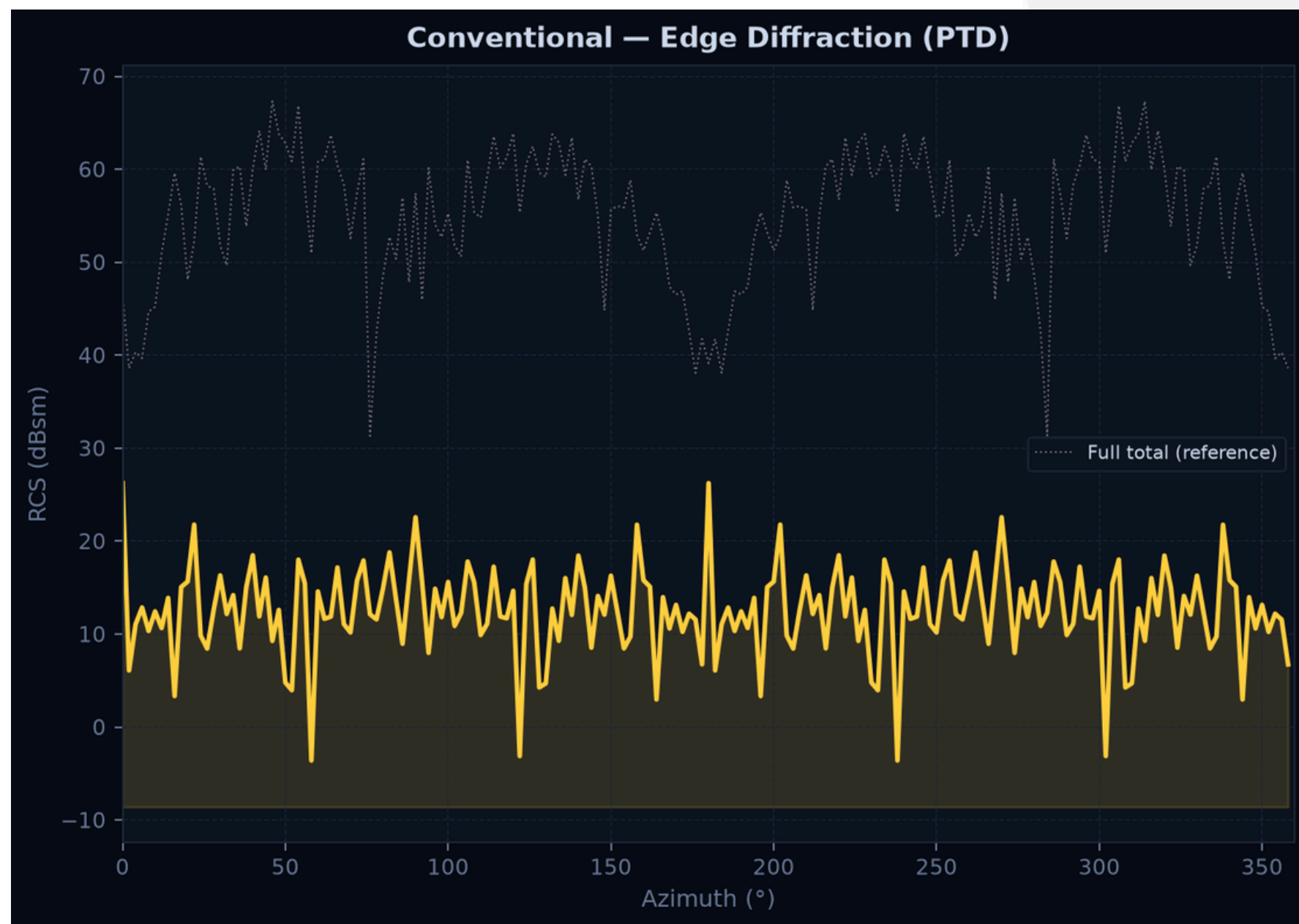
Stealth — Specular Reflection (PO)





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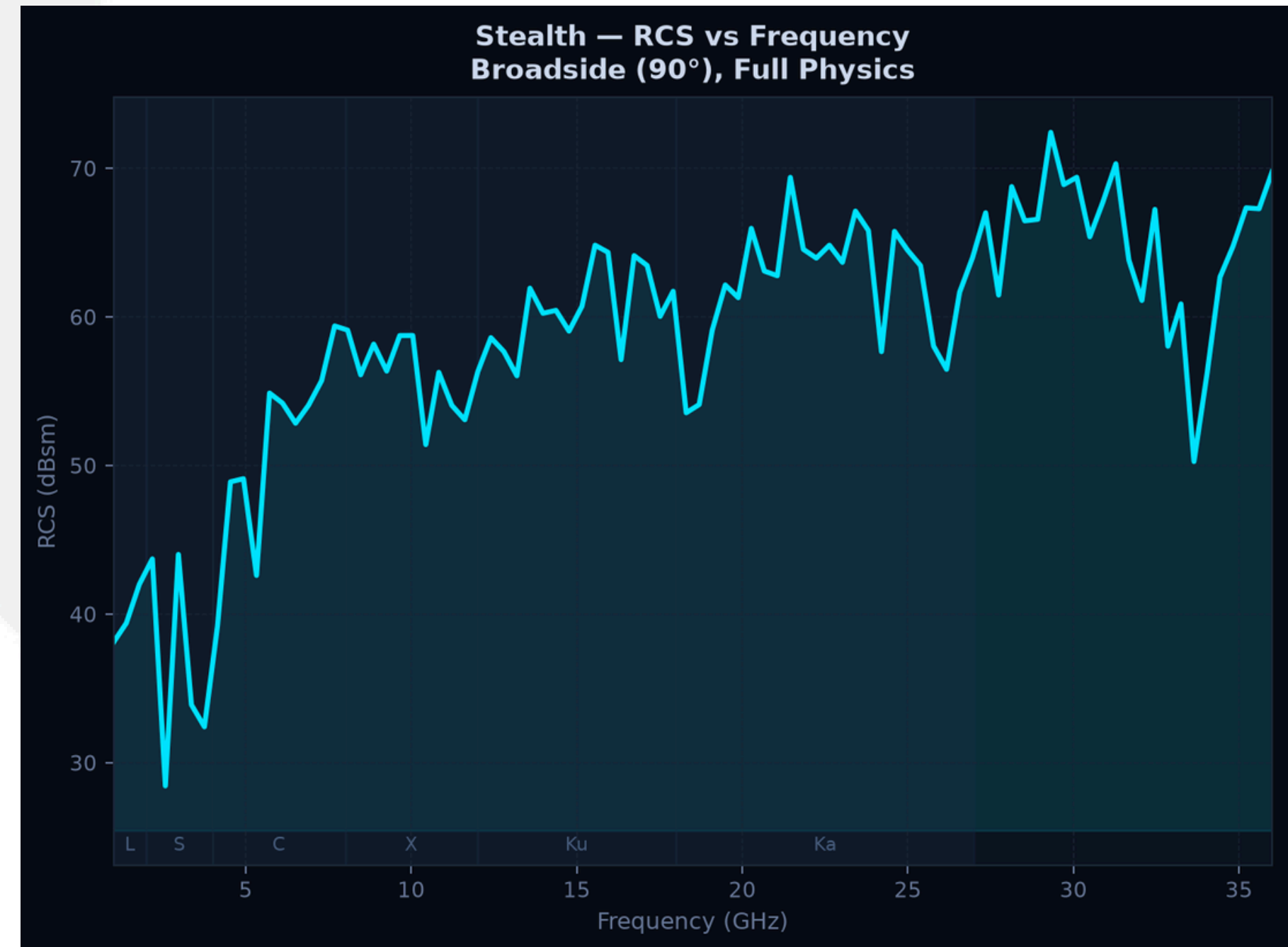
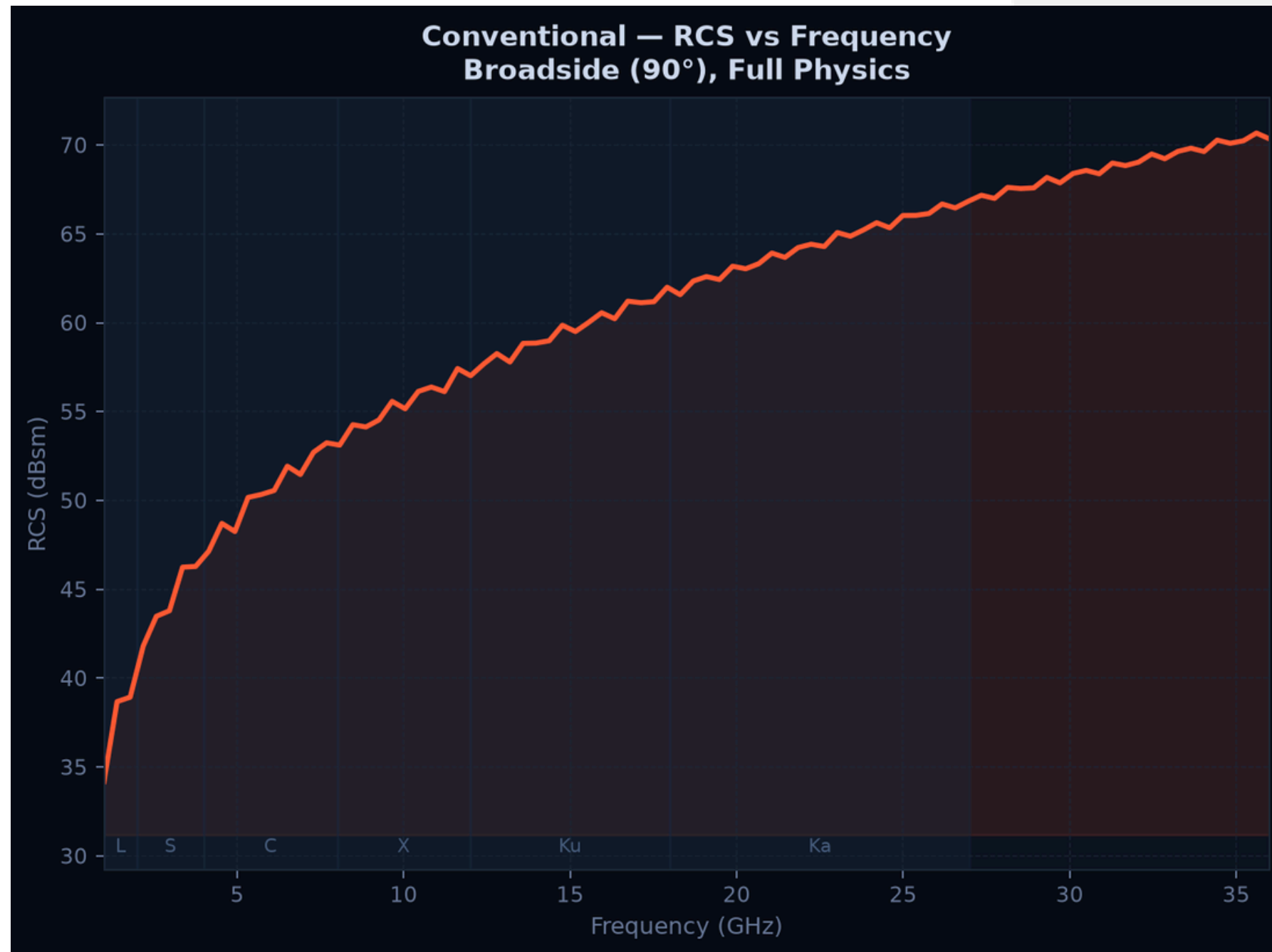




RESULTS - Frequency Sweep:

L → Ka band

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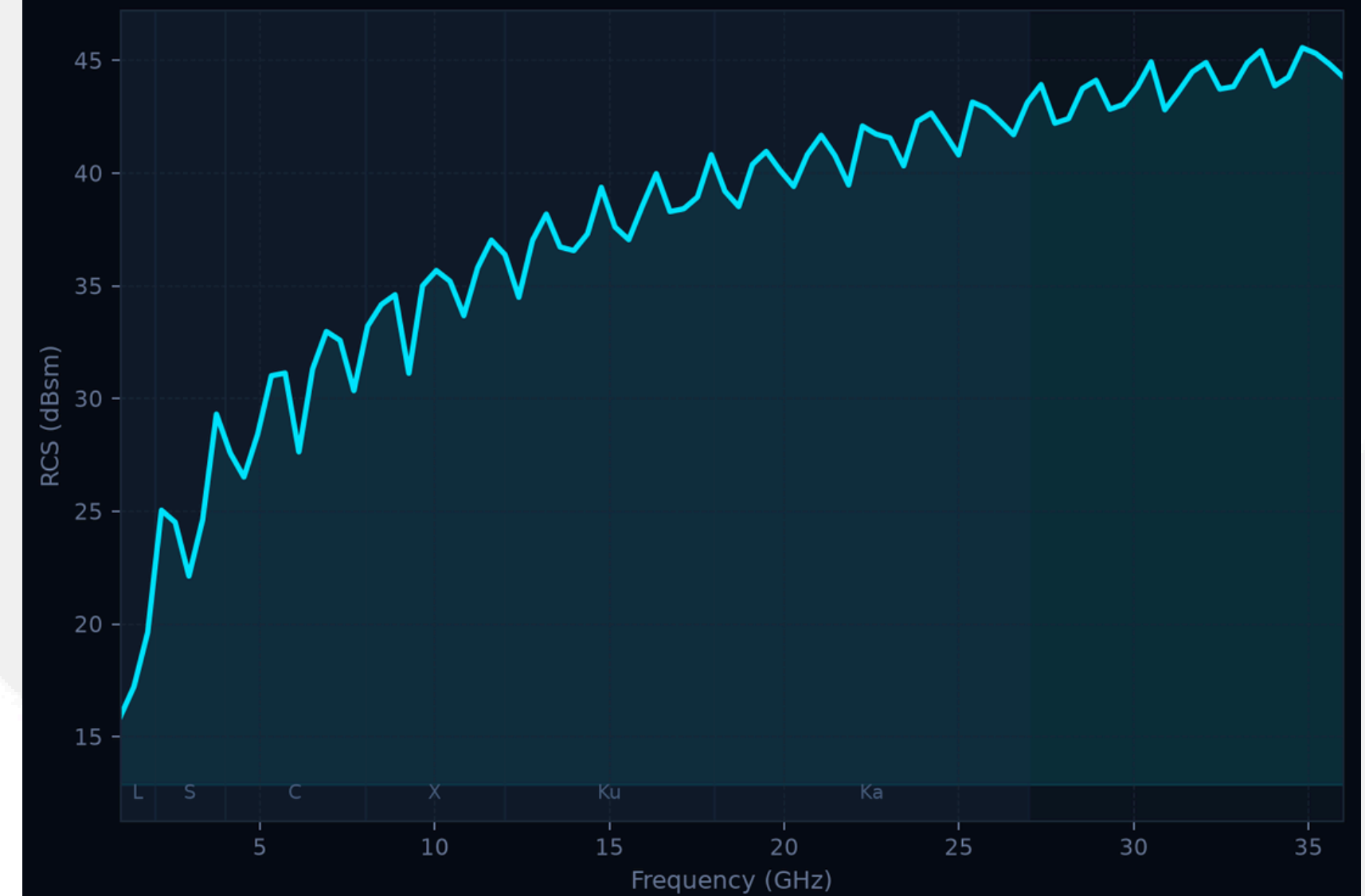




Conventional — RCS vs Frequency
Nose-on (0°), Full Physics



Stealth — RCS vs Frequency
Nose-on (0°), Full Physics



- At L-band (1.5 GHz): stealth is 5.7 dB quieter than conventional at nose-on



Results 2: Geometry Comparison

Frequency Sweep (Nose-on Reduction)

Band	Frequency	Reduction (dB)
L-band	1.5 GHz	5.7
X-band	10 GHz	11.9
Ka-band	24 GHz	11.6

A radar tracking the conventional aircraft at 200 km first detects the stealth aircraft at only 100 km — a 50% reduction in detection range from an 11.9 dB nose-on RCS reduction.



Results 3: Edge and Double- Bounce Analysis

Design Principles

Sweep leading edges

- Mechanism: Sweeping shifts the diffraction peak by angle Λ° off nose-on, moving the return outside the frontal threat corridor for $\Lambda > 60^\circ$.
- Evidence: Stealth aircraft's $\sim 65^\circ$ swept delta wings with aligned trailing edges.

Cant vertical surfaces

- Mechanism: A single design change simultaneously reduces specular return and eliminates the 90° fin-fuselage double-bounce corner.
- Evidence: Stealth aircraft's V-tails canted 35° outward.

Combine geometric shaping with RAM

- Mechanism: Shaping redirects scattered energy away from the threat sector; RAM reduces the amplitude of remaining returns — Salisbury screens give deep narrowband absorption, Dallenbach layers give broader shallower absorption.
- Evidence: Complementary application across design principles 1–5.



CONCLUSION

- **The stealth aircraft achieves 11.9 dB RCS reduction at nose-on over the conventional aircraft equivalent to the radar's detection range dropping from 200 km to 100 km**
- **All three physics layers are essential — double bounce alone accounts for the 45°-peak spikes that PO+PTD completely miss**
- **Edge classification diagnostics confirm the stealth aircraft has zero internal right-angle ridge structure($n=2.0$ uniformly, 44 boundary edges out of 52)**
- **PO implementation validated against Mie series to within 0.2 dBsm**



1	E. F. Knott, J. F. Shaeffer, and M. T. Tuley, Radar Cross Section, 2nd ed. SciTech Publishing, Raleigh, NC, 2004.
2	P. Ya. Ufimtsev, Fundamentals of the Physical Theory of Diffraction. John Wiley & Sons, Hoboken, NJ, 2007.
3	M. I. Skolnik, Introduction to Radar Systems, 3rd ed. McGraw-Hill, New York, 2001.
4	C. A. Balanis, Advanced Engineering Electromagnetics, 2nd ed. John Wiley & Sons, Hoboken, NJ, 2012.
5	R. F. Harrington, Field Computation by Moment Methods. Macmillan, New York, 1968.